

Chapter 39. Complications of Translaryngeal Intubation

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Complications of Translaryngeal Intubation: Introduction

The term *endotracheal intubation*, or *tracheal intubation*, broadly refers to the insertion of a definitive artificial airway into the trachea by either the translaryngeal or transtracheal route. *Translaryngeal intubation* (TLI), a more specific term, is transoral or transnasal intubation of the airway through the larynx. The term *endotracheal tube* (ETT), as opposed to *tracheostomy tube*, refers to a tube passed via the mouth or nose into the trachea.

Macewen is usually given credit for the first successful translaryngeal intubation. In a landmark paper in 1880, he described TLI in four patients for as long as 35 hours and reported complications, including cough, discomfort, tracheal mucosal congestion, and thickening of the vocal cords and posterior rim of the glottis.¹ The modern era of TLI began in the 1950s when it was used to manage respiratory failure from drug overdose^{2,3} and polio,⁴ and to provide an alternative to immediate tracheotomy when patients required prolonged ventilator support. As experience with TLI grew, so did knowledge of its complications and limitations. Comprehensive reviews of this subject appeared first in 1950⁵ and frequently thereafter.^{6–14}

This chapter provides a practical review of complications of TLI in critically ill adult patients, helps the reader understand the mechanisms underlying these adverse events, and illustrates how awareness of these complications may lead to their prevention, earlier recognition, and successful management. The chapter reviews important complications and consequences of TLI with standard ETTs that are encountered in the practice of adult critical care medicine. They are arranged according to the outline in [Table 39-1](#). Complications of anesthetic intubations are occasionally cited to add awareness of complications that may also be seen in the critical care setting.

Table 39-1: Classification of Complications of Translaryngeal Intubation

Temporal classification
Complications during endotracheal tube (ETT) placement
Complications while the ETT is in place
Complications during and after extubation
Anatomic-physiologic classification
Nasal and paranasal
Oral
Pharyngeal
Laryngeal
Tracheal
Bronchial
Pulmonary
Miscellaneous
General
Esophageal
Gastric
Musculoskeletal
Neurologic
Physiologic
Other

Frequency of Complications

The complication rate of TLI varies with the setting, the urgency of the procedure, the skill of the intubator, patient anatomy, and other factors. In the prehospital setting, the rate of complications of TLI ranges from 9.5¹⁵ to 22.7%.¹⁶ In the emergency department, 8%¹⁷ to 38%¹⁸ of patients undergoing TLI

experience one or more complications.

In a prospective study in the adult critical care setting, Stauffer et al observed that 62% of all TLIs had one or more associated adverse events either during intubation or while the ETT was in place.¹⁹ The mean number of complications per patient was 1.2. The most common early clinical problems were, in descending order of frequency, excessive cuff pressure required to seal the airway, self-extubation, inability to seal the airway, right main-stem bronchus intubation, and aspiration (Table 39-2). Other prospective studies have reported that 28%²⁰ to 39%²¹ of TLIs in the intensive care units (ICUs) of academic centers were associated with one or more complications, many of which were life-threatening. The complication rate of emergency TLI by anesthesia residents in ICUs and wards of large tertiary academic health centers ranges from 4.2%²² to 16.1%,²³ with endobronchial intubation, airway trauma, aspiration, and esophageal intubation leading the list of observed complications.

Table 39-2: Early Complications of Translaryngeal Intubation after 226 Intubations in 143 Adults

Clinical Problem	No.	Percentage
Excessive cuff pressure to achieve seal by minimal occluding pressure technique ^a	42	19
Self-extubation	29	13
Inability to seal airway ^b	24	11
Right main-stem bronchus intubation	21	9
Aspiration ^c	17	8
Lip ulceration or cellulitis	16	7
Pharyngeal injury or bleeding	15	7
Mechanical problems with the ETT ^d	14	6
Difficulty suctioning via the ETT	12	5
Pain in nose, mouth, pharynx, or chest related to the ETT	8	4
Glottic edema	5	2
Oral mucous membrane injury	5	2
Tooth avulsion	4	2
Laryngospasm at the time of extubation	3	1
Pneumothorax	2	1
Esophageal intubation	2	1
Miscellaneous ^e	49	22
Total number of complications observed: 268		

^aDefined arbitrarily as >25 mm Hg.

^bDefined as leakage of air around the cuff because of a defect in the cuff or inadequate seal with cuff pressure <60 mm Hg.

^cDefined as aspiration apparent clinically as an immediate result of the intubation attempt or after successful intubation.

^dIncludes partial dislodgment of the tube, length too short or too long, cuff laceration, and biting and occluding the tube.

^eIncludes nasal bleeding, which was recorded twenty-two times.

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Little is known about the impact of TLI complications on utilization of health care resources. One report found a significant increase in health care costs and hospital length of stay resulting from tracheal injury from ETTs.²⁴

Complications during Endotracheal Tube Placement

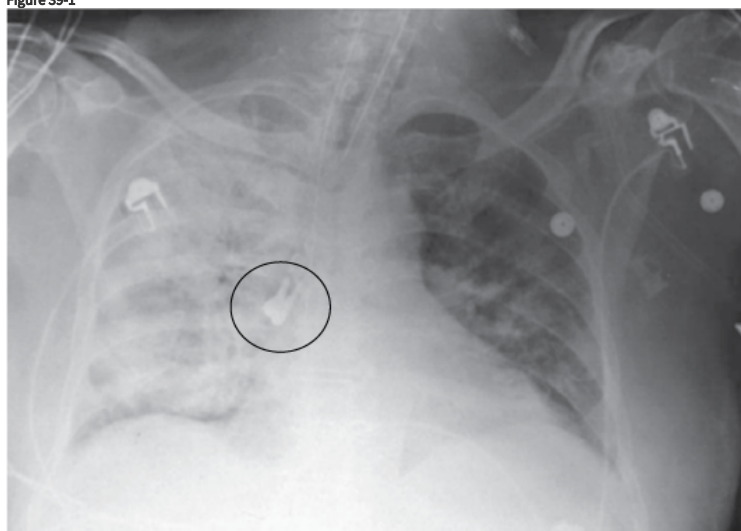
Nasal and Paranasal

Nasal intubation may inflict trauma to the nose, especially when it is performed urgently or without adequate premedication. A retrospective study of 105 nasotracheal intubations in the emergency department revealed a 26% rate of immediate complications, including epistaxis, emesis, and main-stem bronchus intubation, and a 23% rate of late complications such as pneumonia, sinusitis, and sepsis.²⁵ Nasal bleeding complicated 54% of attempts at nasal intubation in an adult ICU.¹⁹ Dislodgment of nasal turbinates²⁶ and nasal polyps²⁷ is rare. Intracranial placement of nasotracheal tubes, particularly in the setting of facial trauma, has also been reported.²⁸

Oral

Overall, dental injury from TLI is uncommon, occurring in approximately 0.9% of prehospital TLIs,²⁹ 0.04%³⁰ to 0.13%³¹ of anesthesia TLIs, and 0.2% of emergent nonoperating room TLIs.²² In anesthetic intubations, loosening, luxation, or avulsion of teeth, damage to crowns and bridges, and crown or root fracture were the most commonly observed dental injuries in two large series.^{30,31} Dental injury is the most common reason for anesthesia-related malpractice claims.³¹ Avulsed teeth and dental appliances may be aspirated into the trachea or bronchi.³² Figure 39-1 illustrates aspiration of a tooth in the setting of endotracheal intubation.³² Difficult intubation, poor dentition, and inexperience of the intubator predispose to dental injury.³⁰ Oral mucous membrane injury has been observed in 1% to 2% of attempts at oral intubation.¹⁹ Lip injury occasionally occurs from pressure exerted by the laryngoscope blade. Temporomandibular joint dislocation is a very rare complication of anesthetic orotracheal intubation.³³

Figure 39-1



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Chest radiograph showing an aspirated molar tooth in the right main bronchus of an intubated patient. Pneumonia in the right lung was also observed. (Reproduced, with permission, from Ostrinsky Y, Cohen Z. Tooth aspiration. *N Engl J Med*. 2006;354:e25. Copyright © Massachusetts Medical Society.)

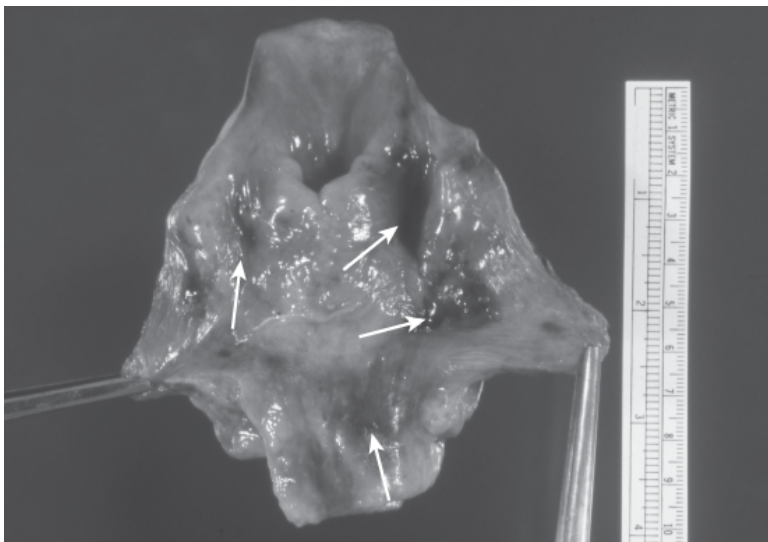
Pharyngeal

Lacerations, bleeding, contusions, excoriations, submucosal hemorrhage, and edema may result from intubation trauma to the nasopharynx, oropharynx, or hypopharynx. Oropharyngeal and hypopharyngeal injuries occur from trauma by the laryngoscope blade, ETT, stylet, or dislodged dental appliances.

Perforation of the posterior pharyngeal wall or hypopharynx is a particularly serious complication. It may result in mediastinal and subcutaneous emphysema, hematoma, upper airway obstruction, abscess formation,^{34,35} mediastinitis, pneumothorax, pneumomediastinum,³⁶ pneumoperitoneum,³⁷ and even cardiac arrest.^{8,38,39}

Hypopharyngeal injury from TLI has been the subject of a number of reports in both adults^{8,37-42} and children.⁴³ Like oral injury, pharyngeal trauma from TLI occurs mainly in emergency settings and in the hands of inexperienced intubators, but it may occur in the anesthetic setting as well.⁴⁴ Piriform sinus laceration by forceful blind intubation may result in severe barotrauma with bilateral pneumothorax, pneumomediastinum, mediastinal abscess, and cardiac arrest.⁴⁰ Figure 39-2 illustrates hypopharyngeal contusions from repeated unsuccessful attempts at oral intubation by an inexperienced intubator.

Figure 39-2



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Autopsy specimen of larynx, hypopharynx, and upper esophagus opened posteriorly. The patient was an older man with chronic obstructive pulmonary disease (COPD) and acute respiratory failure in whom repeated forceful attempts at oral intubation by an inexperienced physician were not successful. The tip of the ETT traumatized the airway. Multiple areas of submucosal hemorrhage in the piriform sinuses and hypopharynx are evident (*arrows*).

Laryngeal

Reports of laryngeal complications that occur during ETT placement come mainly from the anesthesia literature. The larynx is the most common site of airway injury in anesthesia claims analysis.⁴⁵ Laryngeal injuries during ETT placement include glottic contusion, vocal cord hematoma and laceration, vocal process avulsion,⁴⁶ and arytenoid cartilage dislocation.^{46,47} Vocal cord hematomas are reported to be more common on the left true vocal cord,⁴⁸ a finding that has been attributed to the fact that most intubators are right-handed.¹⁰ Epiglottic hematoma from TLI is considered very rare.^{49,50} Arytenoid dislocation from traumatic intubation occurs in up to 1.7%⁴⁷ of anesthetic intubations, leading to complications such as hoarseness and aspiration⁴⁷ and acute respiratory failure.⁵¹ Colice et al observed no cases of this complication in critical care intubations.⁵²

Tracheal

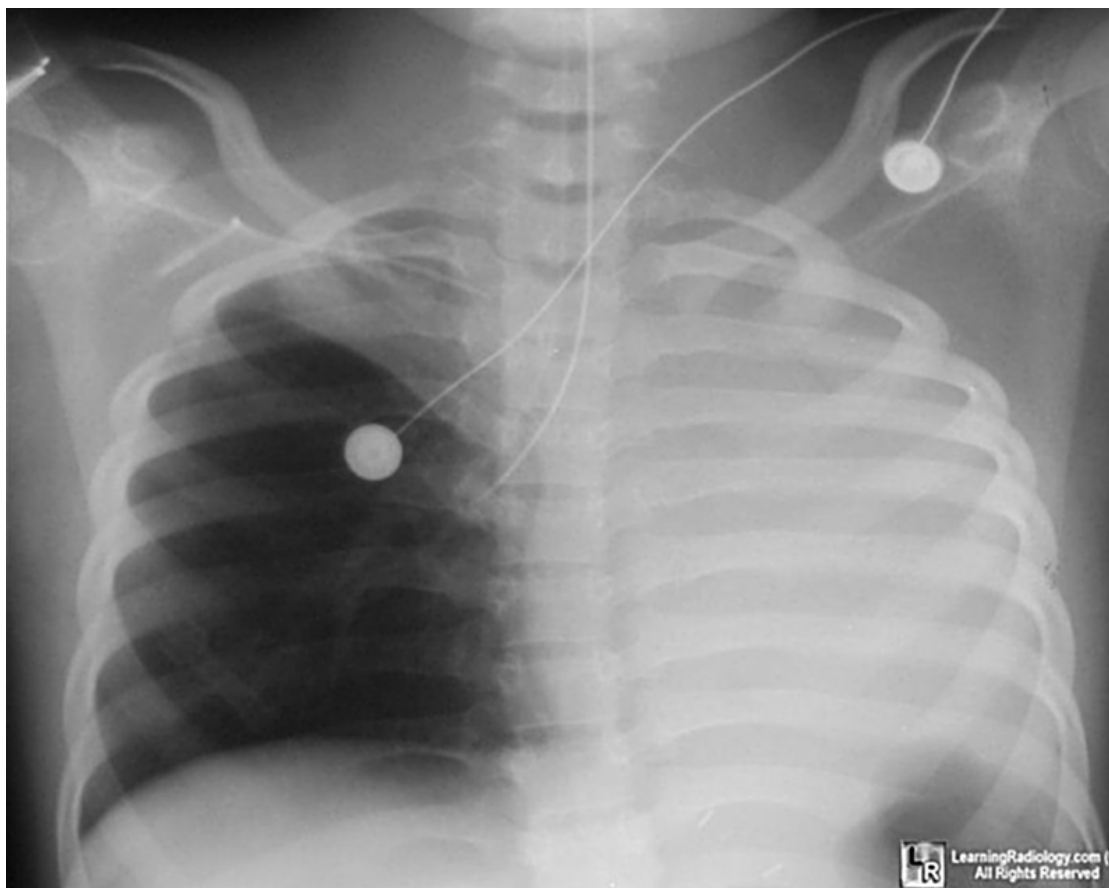
Tracheal tears, reported variously as perforation,⁵³ laceration,⁵⁴ and rupture,^{55–59} are rare complications of TLI. The large majority of cases occur in women and in patients older than 50 years of age,⁶⁰ and involve the posterior membranous portion of the trachea at or near the carina.⁶¹ Causes of tracheal tears include emergency intubation, damage to the posterior membranous trachea by a stylet,⁵⁶ overinflation of the cuff,^{43,56} excessive coughing or movement during intubation, and preexisting abnormalities of the trachea such as tracheomalacia.^{55,60,61} Subcutaneous emphysema, pneumomediastinum, and respiratory distress are commonly seen after this type of tracheal injury.^{55,56}

Malpositioning of the ETT in the trachea is a frequent consequence of emergency TLI in both the prehospital and the intensive care settings. In one report of prehospital airway management in a large urban region, malposition of the ETT was detected in 5.2% of 846 patients with attempted TLI in the field who were transported to a hospital.⁶² In a prospective series of 271 emergency TLIs in critically ill adults, Schwartz et al observed nine (3.3%) cases in which the ETT tip was too high (>6 cm above the carina) and twenty-three (8.5%) cases in which it was too low (<2 cm from the carina), in addition to ten cases (3.7%) of right main-stem bronchus intubation.⁶³ Malpositioning of the ETT was significantly more common in women than in men.

Bronchial

Intubation of a main-stem bronchus is a potentially serious complication of intubation.^{19,64,65} Right main-stem bronchus intubation has been reported in 10.7%⁶⁶ and 15.2%⁶⁷ of prehospital TLIs and in 3.7%⁶³ and 9.6%⁶⁴ of cases in prospective series of critical care intubations. In contrast, only 1.6% of anesthetic TLIs are complicated by endobronchial intubation.⁶⁸ Left main-stem bronchus intubation is very rare.⁶⁹ Potential complications of right main-stem bronchus intubation include hyperinflation of the right lung, right pneumothorax, atelectasis of part or the entire left lung, and impaired gas exchange. **Figure 39-3** illustrates right main-stem bronchus intubation with the complication of atelectasis of the left lung.

Figure 39-3



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Chest radiograph showing the tip of the endotracheal tube in the right main-stem bronchus. This case of right endobronchial intubation is complicated by left lung atelectasis. (Reproduced, with permission, from William Herring MD, www.learningradiology.com. Published with permission from LearningRadiology.com.)

Pulmonary

Pulmonary aspiration, which is usually defined as the appearance of new infiltrates on the chest radiograph immediately after intubation, has been observed in 2% to approximately 6% of emergent intubation procedures.^{20–22,29,70} The prospective investigation of 297 TLIs in critically ill adults reported by Schwartz et al revealed twelve cases (4%) of new unexplained lung infiltrates attributed to aspiration and two cases of pneumothorax (1%).⁷⁰ Aspiration of gastric contents was second only to failed intubation among all major complications of anesthetic airway management in a recent national survey in the United Kingdom.⁷¹ Besides pneumothorax, other types of barotrauma may occasionally be seen. Failed intubation attempts may lead to worsening gas exchange, including severe hypoxemia.⁷² Hypoxemia (oxyhemoglobin saturation <90%) has been reported in 18%⁷² to 21%⁷³ of emergency TLIs. Hypoxemia was observed in 10.5% of TLIs when two or fewer intubation attempts were made, compared with 70% of TLIs requiring three or more attempts.⁷² Bronchospasm was reported in three (0.8%) of 358 prehospital TLIs.²⁹ The bronchospasm response to TLI is rarely severe. Patients with asthma are at the highest risk.

Miscellaneous

General

Without adequate premedication, critically ill patients may experience pain in the upper airway, generalized discomfort, and anxiety during intubation. These important complications are difficult to assess objectively and are usually overlooked in large case reviews in the literature.

Bacteremia is an overlooked but potentially important consequence of both orotracheal and nasotracheal intubation.^{74,75} Staphylococci and streptococci are the most common bacterial isolates.⁷⁵ The frequency of transient bacteremia following orotracheal intubation ranged from 3.2%⁷⁴ in elective TLI for anesthesia to 9% in urgent TLI performed in ICU patients.⁷⁶ In the latter series, only *Streptococcus* species were isolated immediately after TLI, and no patient had streptococcal bacteremia before or 1 hour after TLI. Bacteremia has been detected following 5.5%⁷⁷ to 16%⁷⁸ of nasotracheal intubations.

Esophageal

An analysis of claims for airway injury from anesthesia showed that the esophagus was the third most common site of injury, and esophageal injuries were more severe than all other types of airway injury combined.⁴⁵ Inadvertent perforation of the posterior wall of the esophagus or pharynx by the ETT tip or rigid stylet is a rare but serious complication of TLI. Risk factors for this injury include inexperience of the operator, anatomic abnormalities, and difficult intubation, especially in the setting of cardiopulmonary resuscitation. The literature describing this complication is limited to single case reports^{79,80} and small series. O'Neill et al described two cases of esophageal and pharyngeal perforation from elective anesthetic intubation, noting that this severe complication is not always limited to emergency or difficult intubations.⁴¹ The potential consequences of esophageal perforation include subcutaneous emphysema, pneumomediastinum, pneumothorax, mediastinitis, and mediastinal abscess. Anterior esophageal perforation from TLI has not been reported.

Esophageal intubation is a common complication of TLI, particularly in emergency settings, and it has potentially severe consequences, including hypoxemia, regurgitation and aspiration of gastric contents, cardiac arrest, and irreversible hypoxic brain injury.^{81–83} Esophageal intubation occurred in ten (6.7%) of 149 patients intubated in the prehospital setting in Germany⁶⁶ and in thirty-seven (5.4%) of 691 patients intubated in the prehospital setting in France.²⁹ It was reported in thirty-three (5.5%) of 603 patients intubated in an emergency department.¹⁷ In tertiary care centers the frequency of esophageal intubation complicating emergency TLI ranges from 1.3% to 9.7%.^{22,67,70,72}

Gastric

Gastric distension may result from esophageal intubation. Swallowing of foreign objects such as teeth and dental fixtures may also occur.⁸⁴ TLI may stimulate the gag reflex, leading to vomiting.

Musculoskeletal

Musculoskeletal complications during intubation are uncommon. Injuries to the cervical spine and cervical spinal cord during orotracheal intubation have been reported.⁸⁵

Neurologic

Hypoxic brain injury may result from ineffective ventilation, oxygenation, or circulation as a result of failed intubation or mishaps such as esophageal intubation. Cervical spinal cord injury is reportedly a rare complication of anesthetic TLI.⁸⁶ Neck hyperextension during direct laryngoscopy to facilitate TLI in patients with restricted neck mobility or preexisting cervical spine instability has been proposed as a risk factor for cord injury,⁸⁷ but proof of this relationship has been difficult to establish.⁸⁸ Transient increases in intracranial pressure occur with TLI, particularly with endotracheal suctioning,⁸⁹ putting patients with head injury at risk for secondary brain injury.⁹⁰

Physiologic

Cardiovascular

The cardiovascular pressor response to laryngoscopy and TLI includes transient and variable increases in heart rate and both systolic and diastolic blood pressure as a result of pharyngeal stimulation. The magnitude of these responses varies with the route and ease of TLI,⁹¹ the use of anesthetic agents, concurrent medications, underlying cardiovascular status, and many other factors. In healthy individuals these cardiovascular responses are usually inconsequential, but complications may result when they are exaggerated⁹² or occur in patients with hypertension, cardiac disease, or cerebrovascular disease.⁹³ Some, but not all, studies document a rise in plasma epinephrine and norepinephrine levels with TLI.^{94–97}

The hypertensive response to nasotracheal intubation is greater than that to orotracheal intubation.⁹⁶ The hemodynamic response to orotracheal intubation is greater than that to insertion of the laryngeal-mask airway (LMA North America, San Diego, CA),⁹⁴ but less than that to insertion of the esophageal-tracheal Combitube™ (Covidien, Mansfield, MA).⁹⁴

Hypotension may also occur during TLI. In the setting of rapid sequence intubation for general anesthesia, Istvan observed hypotension in 27% of 248 patients undergoing appendectomy.⁹⁸ Franklin et al observed that twenty-four (29%) of eighty-four patients requiring TLI in an emergency department developed life-threatening hypotension, which was significantly associated with the presence of chronic obstructive pulmonary disease, hypercapnia, and hypoxemic respiratory failure, but not with the administration of sedatives or paralyzing medications.⁹⁹ Griesdale et al observed severe hypotension in thirteen (9.6%) of 136 patients requiring TLI in an academic ICU.²¹

MacKenzie et al reported cardiac arrhythmias in 58% and 32% of nasal and oral anesthetic intubations, respectively.¹⁰⁰ Bradycardia was observed in 1.6% of emergency intubations in a large tertiary hospital when two or fewer intubation attempts were required versus 18.5% of intubations requiring three or more

attempts.⁷² Most cases with bradycardia were associated with severe hypoxemia (arterial oxyhemoglobin saturation [SpO₂] <70%) and half culminated in cardiac arrest.⁷²

Other

TLI causes transient but significant increases in intraocular pressure.^{101,102}

Cardiac Arrest and Mortality Rate of Translaryngeal Intubation

Cardiac arrest from emergency TLI is common. Mort reported cardiac arrest in sixty (2%) of 3035 critical care patients during emergency TLI in a tertiary care institution. Inadvertent esophageal intubation with profound hypoxemia, regurgitation, aspiration, and bradycardia were highly associated risk factors for cardiac arrest.¹⁰³ Adnet described cardiac arrest as a complication of TLI in four (1.1%) of 358 patients with initial conditions other than cardiac arrest who required intubation in the prehospital setting.²⁹

Schwartz et al reported a mortality rate of 0.85% during or within 30 minutes of emergency TLI in critical care patients without preexisting hypotension.⁷⁰ Biboulet et al prospectively recorded eleven anesthesia-related cardiac arrests in a series of 101,769 anesthetics performed from 1989 to 1995 (frequency: 1.1 per 10,000), and the mortality rate was 0.6 per 10,000.¹⁰⁴ All the cardiac arrests were considered avoidable.

Complications While the Endotracheal Tube Is in Place

Nasal and Paranasal

Nasal Complications

In a prospective series of 379 patients with nasotracheal intubation, Holdgaard et al found inflammation and ulceration of the nostrils or nasal septum in 110 patients (29%) within 5 days of extubation.¹⁰⁵ Nasal bleeding occurred in 19% of patients and conchae fractures in 11%. Necrosis of the nasal alae may occur at the site of constant pressure on the nose by an upturned nasotracheal tube (Fig. 39-4).¹⁰⁶ This injury was seen in 4% of nasotracheal intubations,¹⁰⁷ and it may be accompanied by bacterial invasion with cellulitis.¹⁰⁸ Nasal septal ulceration is uncommonly seen.¹⁹

Figure 39-4



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Hemorrhagic ulceration of the left ala nasi (arrow) as a result of pressure exerted by a previously upturned nasotracheal tube. This type of injury can be prevented by positioning the endotracheal tube so that its proximal end points straight down from the anterior nares. (Modified, with permission, from Zwillich and Pierson.¹⁰⁶)

Paranasal Complications

Sinus Effusions

Nasotracheal intubation for more than a few days commonly results in accumulation of fluid in the paranasal sinuses.^{108–110} Fassoulaki and Pamouktsoglou found computed tomographic (CT) scan evidence of paranasal sinus fluid accumulation, opacification, or mucosal thickening in all of sixteen adult patients

studied prospectively by the eighth day of nasotracheal intubation.¹⁰⁹ The maxillary and sphenoid sinuses each were affected in 87% of cases, followed by the ethmoid (50%) and frontal (12.5%) sinuses. Sinus effusions were unilateral and on the same side as the nasotracheal tube. Involvement of more than one sinus at a time was common.¹⁰⁹ Ultrasound evaluation has demonstrated sinus effusions in 30% of patients with nasotracheal intubation¹⁰⁸ and 63% of patients with orotracheal intubation.¹¹⁰

Sinusitis

In 1982, Knodel and Beekman called attention to nasotracheal intubation as a cause of maxillary sinusitis with fever in mechanically ventilated patients.¹¹¹ Sinusitis has subsequently been recognized as a common and potentially serious complication of TLI, particularly nasotracheal intubation (see [Chapter 47](#)).^{109,110,112–114} Sinusitis may also complicate orotracheal intubation¹¹⁵ and the use of nasogastric tubes.

Severe complications of TLI-associated sinusitis include fever, bacteremia, pneumonia, sepsis, and meningitis. Van Zanten et al reported that sinusitis was discovered as the sole cause of fever in fifty-seven (16.2%) of 351 critically ill patients with orotracheal intubation who were evaluated for fever of unknown origin.¹¹⁵ Gram-negative bacteria such as *Klebsiella* and *Enterobacter* species are commonly isolated from sinus aspirates in patients with sinusitis complicating nasotracheal intubation. Gram-positive bacteria, particularly *Staphylococcus aureus*, and fungi are also frequently isolated. Anaerobic infection, particularly with *Bacteroides* species, also occurs. Polymicrobial infections are common.

Middle Ear Effusions and Otitis

Middle ear effusion is an underrecognized but common complication of prolonged TLI, occurring in as many as half of patients.^{116,117} Cavaliere et al observed that twenty-eight (80%) of thirty-five unconscious ICU patients had middle ear effusion, and there was a significant association with TLI and mechanical ventilation.¹¹⁸ Lucks et al found middle ear effusions in twenty-three (29%) of seventy-eight adults with prolonged TLI, most of whom were intubated orally.¹¹⁹ Tympanocentesis revealed organisms in 22% of patients with middle ear effusion, particularly *Pseudomonas aeruginosa*, *Klebsiella oxytoca*, and *Enterobacter cloacae*, typical of the gram-negative organisms colonizing the airway in patients with TLI.¹¹⁸

Oral

Ulceration or cellulitis of the lips was noted in 7% of TLIs in critically ill adults.¹⁹ Lip edema, bleeding, and hemorrhagic crusts are very common during TLI as a result of pressure injury and abrasion by the orotracheal tube, oropharyngeal airway, tape, orogastric tube, or ETT holder.^{19,120} [Figure 39-5](#) illustrates lip injury from prolonged orotracheal intubation.¹²⁰ Hanley et al reported reactivation of herpes simplex virus infections in 53% of patients with oral TLI longer than 48 hours, but clinical findings were noted in only half of these cases.¹²¹ [Figure 39-6](#) demonstrates herpes simplex virus type 1 complicating orotracheal intubation.¹²² Dental injury, stomatitis, erosion of the hard palate, and soft-tissue injury to the tongue and oral mucosa may develop during TLI, and in some cases these complications are severe ([Fig. 39-7](#)).^{19,123} They are reported to be more likely to occur in patients with concurrent use of oropharyngeal airways or those with rigid contracture of the jaw or propulsive movements of the tongue.¹²³

Figure 39-5



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Ulceration of the right side of the upper and lower lip in a patient with prolonged orotracheal intubation. This type of injury can be prevented by proper securing of the endotracheal tube (ETT), minimizing the direct pressure transmitted to the mouth from the ETT and ventilator tubing, and regular oral examination during the period of translaryngeal intubation. (Modified, with permission, from Stauffer.¹²⁰)

Figure 39-6



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Severe hemorrhagic lesions on the lips of a patient with an oral endotracheal tube (ETT) in place. Herpes simplex virus type 1 (HSV-1) was identified by culture and polymerase chain reaction assay. Reactivation of HSV-1 was considered to be related to immunosuppression of critical illness and friction trauma of the lips by the ETT. (Reproduced, with permission, from Tang JW, Chan PKS. Herpes labialis. *N Engl J Med*. 2007;357:1855. Copyright © Massachusetts Medical Society.)

Figure 39-7



A

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Severe injury to the mouth may occur during translaryngeal intubation because of pressure-induced necrosis. This patient was a young man who was intubated because of status epilepticus and need to control the airway. Rigid contracture of the jaw occurred repeatedly while an oral pharyngeal airway was in place to prevent occlusion of the endotracheal tube by biting. Evidence of oral injury included erosion of the hard palate (*arrow*) by the oropharyngeal airway (A), and complete transection of the tongue, photographed at the time of surgical repair (B). (Modified, with permission, from Stauffer and Petty.¹²³)

Pharyngeal

Pharyngeal complications while an ETT is in place are uncommon.

Laryngeal

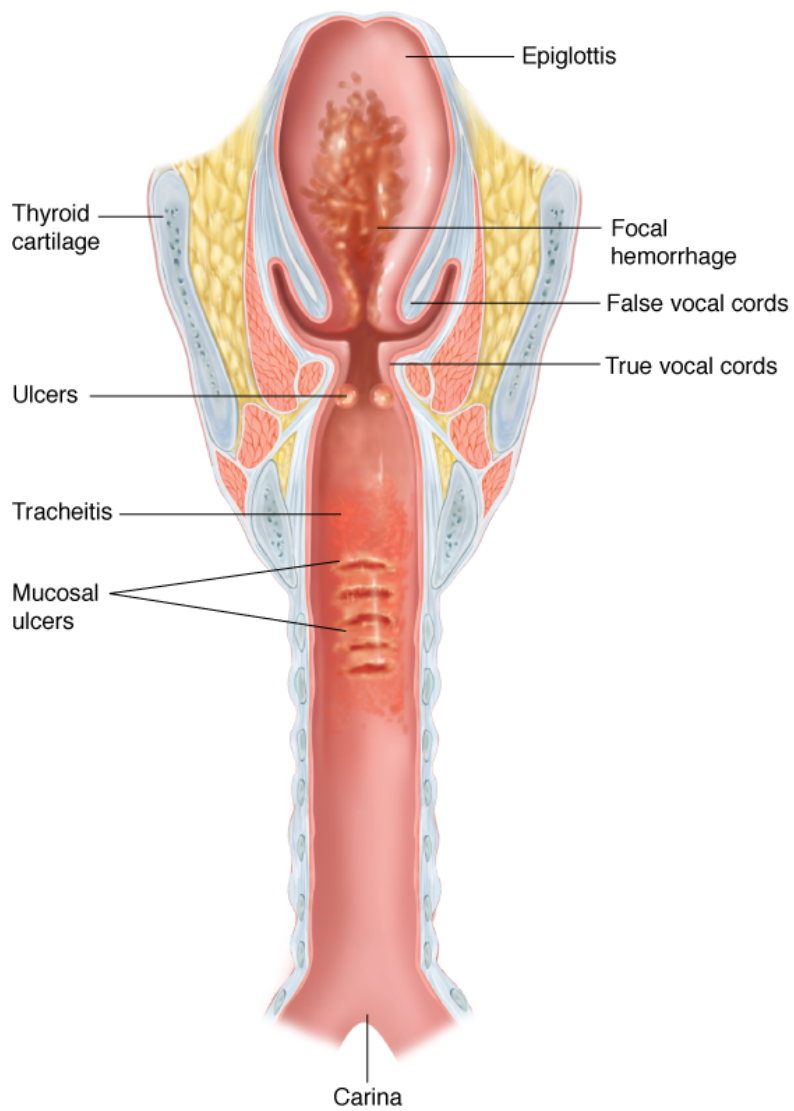
Laryngeal injury is the most common and potentially the most severe complication of TLI,^{52,124–134} and fear of late laryngeal complications of TLI is cited as the primary reason for performing tracheotomy for long-term airway maintenance. The most common postmortem findings in the larynx after prolonged TLI are mucosal ulceration, edema, and submucosal hemorrhage (Table 39-3).^{19,135} Figure 39-8 presents a schematic and illustrations of common TLI-induced laryngotracheal lesions. Figure 39-9 presents endoscopic views of selected cases of glottic injury from TLI in Lindholm's comprehensive study.¹³⁰

Table 39-3: Common Autopsy Findings after Prolonged Translaryngeal Intubation in Forty-One Patients

			Mucosal			
	Mucosal Ulceration		Inflammation and/or Edema		Submucosal Hemorrhage	
Site	No.	%	No.	%	No.	%
Epiglottis	5	12	3	7	2	5
Glottis	21	51	12	29	5	12
Subglottis	5	12	1	2	3	7
Trachea, cuff site	6	15	16	39	3	7
Trachea, other site	2	5	20	49	3	7
Total	39	–	52	–	16	–

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Figure 39-8



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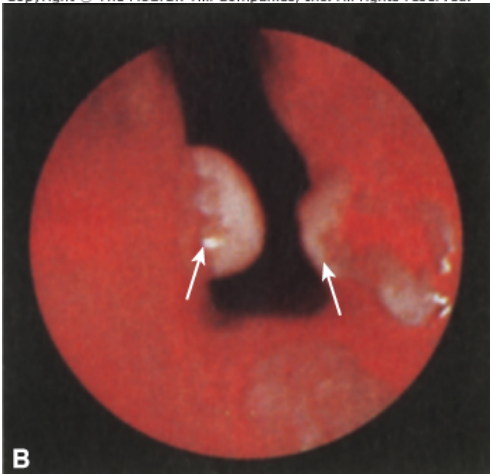
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Schematic of the larynx and trachea opened posteriorly, illustrating common laryngotracheal lesions following translaryngeal intubation in the critical care setting (*left*). Autopsy specimen of larynx and trachea from an orally intubated patient showing hemorrhagic ulcers on the posterior aspect of the true vocal cords (*arrows*), patchy areas of mucosal hemorrhage on the anterior wall of the trachea from the subglottis to the carina (*arrows*), and a superficial mucosal ulcer at the site of the endotracheal tube cuff (*arrows*) (*right*). (Left panel modified, with permission, from Stauffer.¹³⁵)

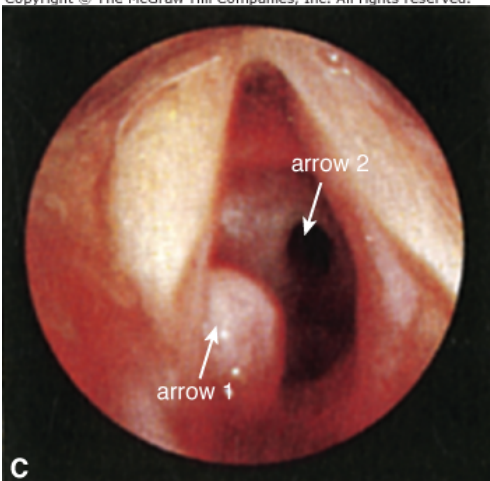
Figure 39-9



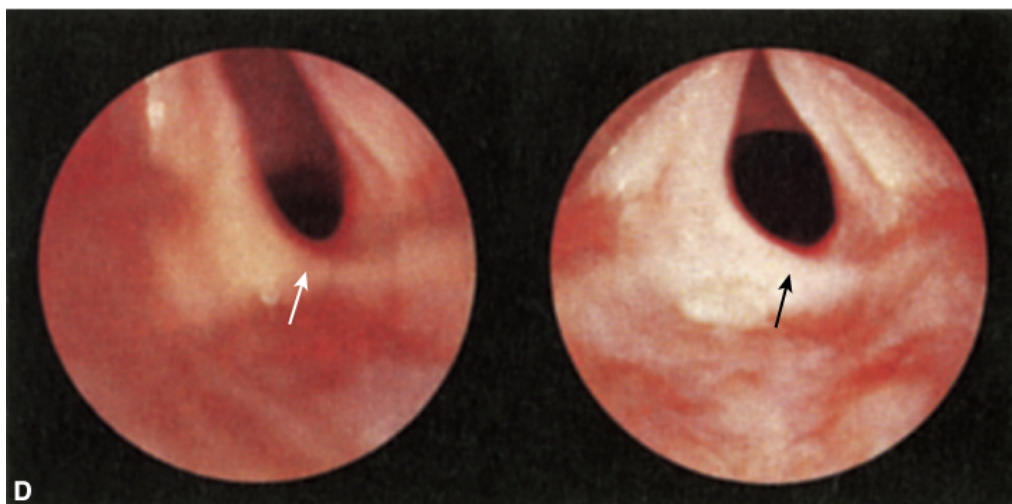
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Endoscopic views representing types of translaryngeal intubation (TLI)-induced laryngeal injury observed by Lindholm in his prospective study of complications of endotracheal intubation.¹³⁰ **A**, Superficial posterior glottic ulcers (*left*) (*arrows*) with healing over 18 days (*center*) and 32 days (*right*). **B**, Bilateral granulomas (*arrows*) that were absent at extubation but appeared by 44 days after extubation. **C**, Left subglottic granuloma (*arrow 1*) 14 months after TLI and tracheotomy. Tracheal stenosis (*arrow 2*) attributed to the tracheotomy is also evident. **D**, Fibrous scar at the left arytenoid region (*arrows*) 15 months (*left*) and 32 months (*right*) after TLI. (Modified, with permission, from Lindholm.¹³⁰)

Experience with prolonged TLI in the 1970s showed moderate to severe laryngeal injury in 4.2%¹³⁶ to 7.2%¹³⁷ of cases. Kambic and Radsel found severe laryngeal injury in sixty-two (6.2%) of 1000 patients after anesthetic intubations, indicating that TLI for short periods and in a controlled setting was not without risk.¹²⁴ Most prospective studies reveal that serious permanent laryngeal injury from TLI is very uncommon.^{19,129,138} All but a few TLI-induced laryngeal injuries tend to heal within 6 to 8 weeks and leave no permanent sequelae.^{52,128,138,139}

Supraglottic Injury

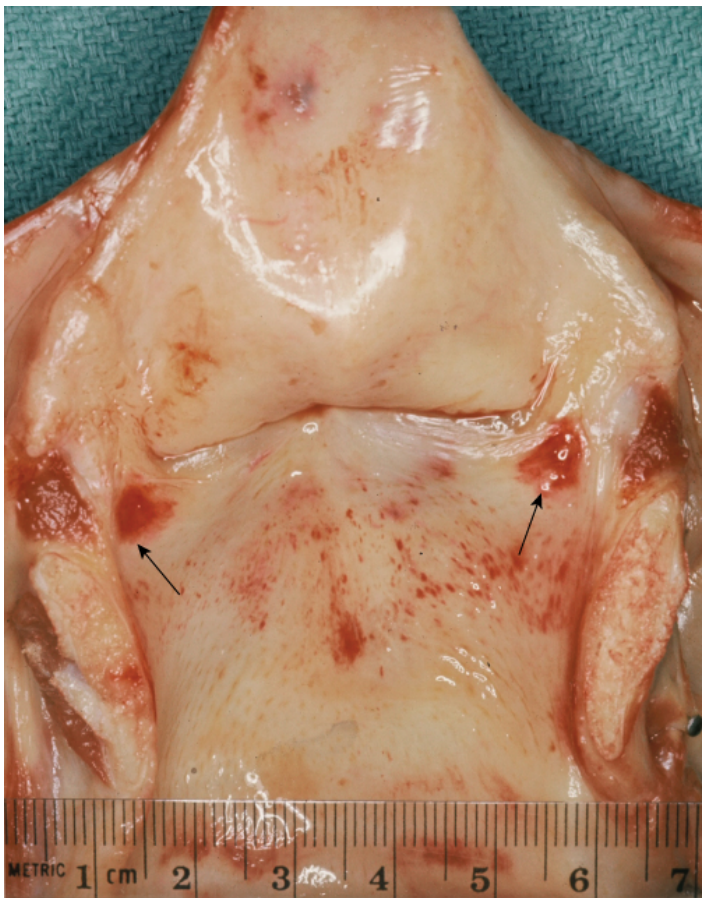
Mucosal ulceration, significant inflammation or edema, and submucosal hemorrhage at autopsy have been noted in 12%, 7%, and 5%, respectively, of patients with prolonged TLI (see Table 39-3).¹⁹ Astrachan et al found severe supraglottic edema in 1% of TLIs.¹⁰⁷

Glottic Injury

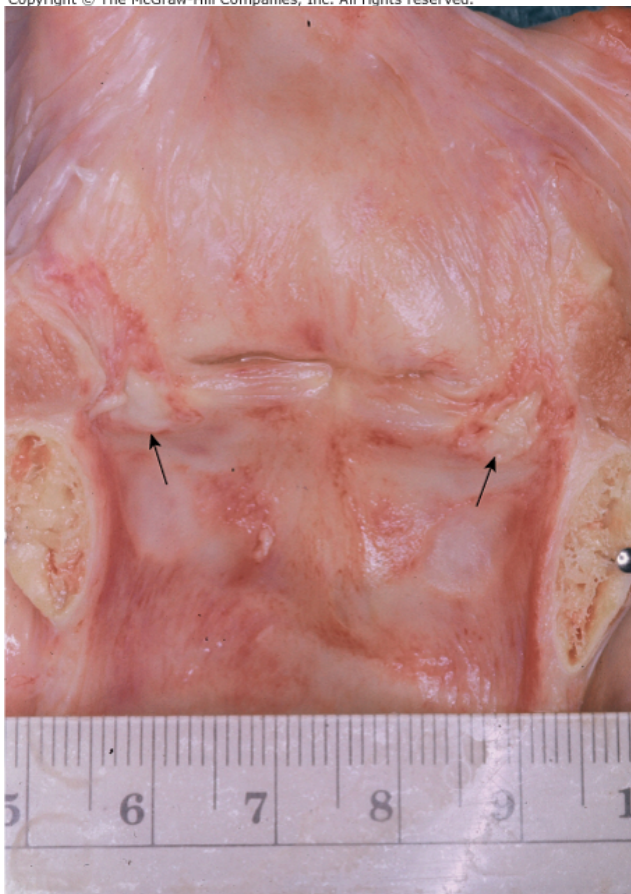
Laryngeal Ulceration

Ulceration of the posterior aspect of the true vocal cords and arytenoids has been reported in 51%¹⁹ to 79%¹³⁰ of TLIs. This injury represents the most common significant complication of TLI.^{47,108,131-134,136,139,140} Investigators first reported posterior laryngeal ulcers from TLI in the early 1950s.^{141,142} Lindholm observed postmortem macroscopic ulceration or necrosis in the interarytenoid area, the medial side of the arytenoids, and the inner posterolateral area of the cricoid cartilage in 9%, 79%, and 68% of adults, respectively.¹³⁰ Laryngeal ulcers from TLI are symmetrical, triangular-shaped erosions of the posterior and medial aspects of the vocal processes (Fig. 39-10) and arytenoids and the posterolateral area of the cricoid cartilages.⁴⁷ They are typically 4 to 12 mm in diameter and 1 to 5 mm in depth, and they may penetrate into the cartilage¹³¹ and the cricoarytenoid joints. The areas most vulnerable to ulceration by pressure from the shaft of the ETT are the posterior endolarynx, the arytenoids, the cricoarytenoid joints, and the interarytenoid space.¹²⁶ Using telarlaryngoscopy, Deeb et al examined 142 adults who required tracheotomy because of failed extubation.¹⁴³ They observed that mucosal erosions and ulcerations, often with exposed cartilage, were common, and the most severe ulcers affected the intercartilaginous glottis and posterior subglottis, often with extensive granulation tissue. In some cases, the paired posterior ulcers may be large enough to join each other, producing a “horseshoe lesion.”¹⁰⁸

Figure 39-10



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Autopsy specimens of larynx from two patients showing typical translaryngeal intubation-induced ulcers on the posterior true vocal cords. *Left*: Superficial hemorrhagic ulcers (*arrows*). Focal mucosal hemorrhage is also seen. *Right*: Deeper ulcers to the level of cartilage (*arrows*).

Glottic Edema and Inflammation

Varying degrees of hyperemia, edema, and inflammation of the glottic mucosa occur in practically all patients during TLI.^{19,52,125,130,139,140} Stauffer et al found severe edema and inflammation at autopsy in 29% of patients following TLI, particularly in the posterior commissure area.¹⁹ Figure 39-11 illustrates an example of intubation-related laryngeal edema.¹⁴⁴

Figure 39-11



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Laryngoscopic view of edema of the vocal folds following translaryngeal intubation. Some patients with this complication, which presents as postextubation stridor, may require reintubation. (Used, with permission, from Wittekamp.¹⁴⁴)

Vocal Cord Paresis and Paralysis

This complication during TLI presents after extubation and is discussed (see the section [Early Complications After Extubation/Laryngeal/Vocal Cord Paresis and Paralysis](#)).

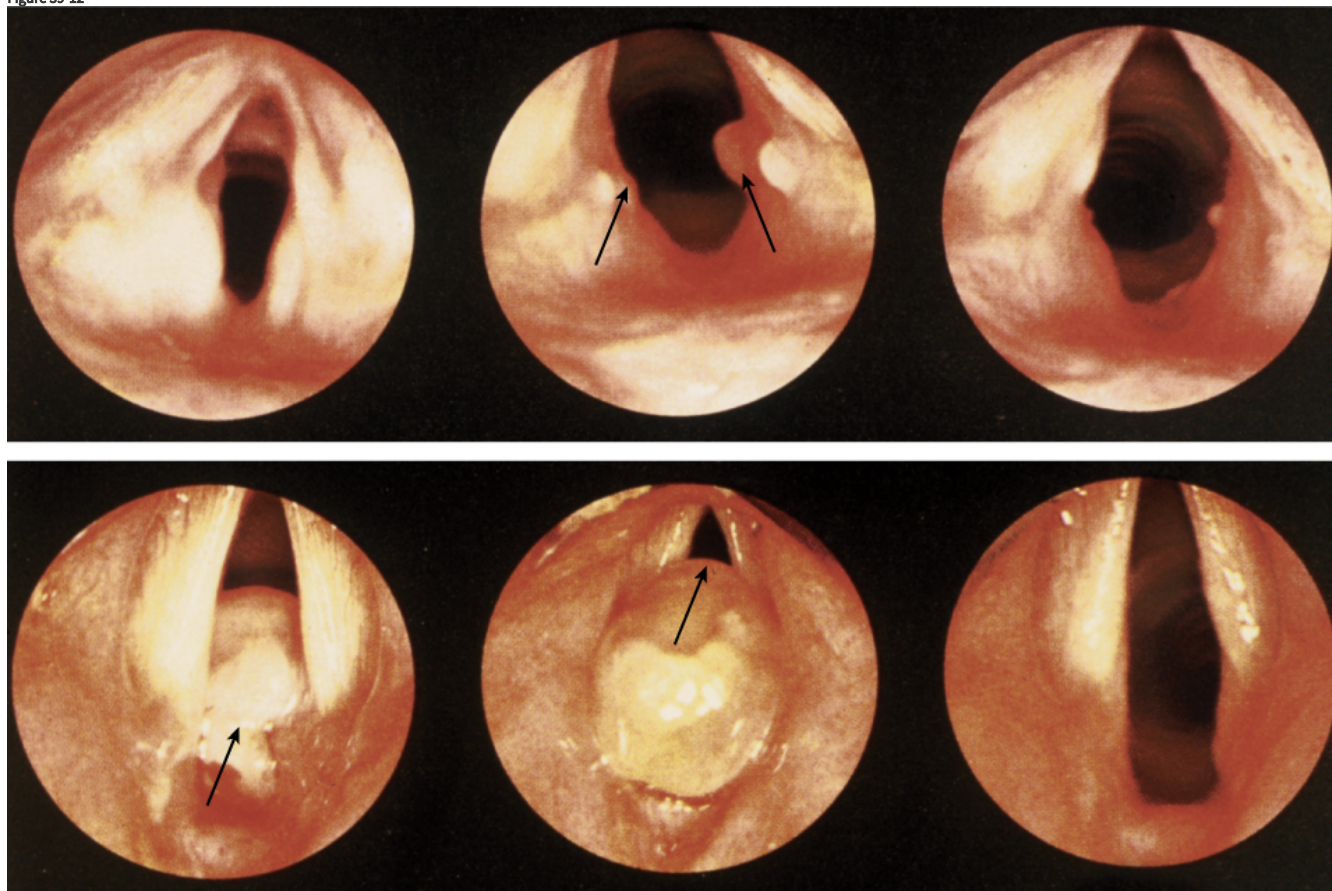
Glottic Hemorrhage

The incidence of glottic hemorrhage developing after intubation is not known. Stauffer et al found evidence of submucosal hemorrhage in the glottis in 12% of TLI cases at autopsy (see [Table 39-3](#)).¹⁹

Granuloma Formation

The term *vocal fold granuloma* refers to growths of granulation tissue at the site of mucosal injury as a result of abnormal tissue healing. These lesions typically appear at the site of laryngeal ulceration and vary in size from 2 to 3 mm to more than 1 cm in diameter. They may be unilateral or bilateral.¹³⁰ Figures 39-9B and C and 39-12 from the series of Lindholm et al,¹³⁰ present endoscopic views of small and large laryngeal granulomas following extubation. Laryngeal granulomas during TLI have been observed in up to 16%¹²⁸ of patients during TLI. Although usually associated with prolonged TLI, laryngeal granulomas may also occur after short-term intubation.^{145,146} This topic is discussed in “Late Complications After Extubation.”

Figure 39-12



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Laryngeal granulomas are growths of granulation tissue that may result from abnormal wound healing at the site of laryngeal ulcers. The *top panel* shows endoscopic views of the larynx 24 hours after extubation (*left view*), 20 days after extubation (*middle view*), and 34 days after extubation (*right view*). Small laryngeal granulomas (one on the left vocal fold and two on the right vocal fold) are evident in the middle view, and marked spontaneous improvement is observed 14 days later (*right view*). The *bottom panel* shows a very large polypoid laryngeal granuloma 5 months following short-term intubation (*left and middle views*) that caused respiratory difficulty and required surgical removal. Healing was achieved following surgery (*right view*). (Used, with permission, from Lindholm.¹³⁰)

Subglottic Injury

Cricoid cartilage abscess results from bacterial invasion of the cricoid cartilage after pressure ulceration by the ETT.^{10,108} Cricoid cartilage abscess may appear as long as 8 weeks after extubation and predispose to posterior glottic stenosis.¹⁰

Tracheal

Tracheal injury during TLI (Table 39-4) may occur at the site of the inflated cuff, at the level of the tip of the tracheal tube, and at the site of injury by suction catheters. Cuff-site injuries are the most common.¹⁹

Table 39-4: Tracheal Complications While the Endotracheal Tube Is in Place

Edema and inflammation
Ulceration of the mucosa and tracheal wall
Granuloma formation
Submucosal hemorrhage
Necrosis
Destruction of cartilage
Tracheal rupture and tracheal laceration
Tracheal dilation
Tracheomalacia
Tracheoesophageal fistula
Tracheoarterial fistula
Epithelial damage
Squamous metaplasia of tracheal epithelium
Reduction in mucociliary clearance
Airway colonization with bacteria
Pseudomembranous tracheitis
Miscellaneous
Too high or too low endotracheal tube position
Tracheobronchitis
Carina irritation, leading to cough and bucking
Suctioning complications

Edema and Inflammation

Fiberoptic bronchoscopy during TLI often reveals varying degrees of tracheal inflammation and edema below the tip of the ETT. Mucosal edema or inflammation is found at autopsy in 39% of TLI patients at the cuff site and in 49% at other levels of the trachea (see [Table 39-3](#)).¹⁹ Infiltrates of neutrophils and mononuclear cells are found in the mucosa and submucosa at the cuff site.¹⁴⁷

Tracheal Ulceration

The use of high-volume, low-pressure cuffs on ETTs has greatly reduced the frequency of cuff-site ulceration. [Figure 39-8 \(right panel\)](#) illustrates a small cuff-site ulcer with exposed cartilage on the anterior tracheal wall. Severe cuff-site ulcers that were common in the era of hard-cuff ETTs^{148,149} are now rarely seen. Modern soft-cuff ETTs, however, do not completely protect the mucosa from injury. Stauffer et al reported tracheal mucosal ulceration at autopsy in 15% of patients with prolonged TLI with soft-cuff ETTs at the cuff site and in 5% at other tracheal sites (see [Table 39-3](#)).¹⁹ Soft cuffs apply lateral pressure over a longer segment than hard cuffs, and pressure from overlapping folds of the fabric of soft cuffs may create channels in the tracheal mucosa.¹⁵⁰

Granuloma Formation

Granulomas in the trachea are much less common than in the larynx following prolonged TLI, because tracheal ulceration is now much less common than laryngeal ulceration. All of the laryngotracheal granulomas observed by Lindholm were in the larynx.¹³⁰ Astrachan et al found tracheal granulomas in only 1% of patients.¹⁰⁷

Submucosal Hemorrhage

Submucosal hemorrhage was found at autopsy in 7% of patients at both the cuff and other tracheal sites (see [Table 39-3](#)).¹⁹

Tracheal Necrosis

Abbey et al reported a case of massive tracheal necrosis and suppuration after 10 days of TLI.¹⁵¹ Possible risk factors include sepsis, hypotension, and elevated cuff pressures.

Destruction of Tracheal Cartilage

Ulceration and eventual destruction of the C-shaped cartilaginous tracheal rings may occur as a result of sustained elevation of cuff pressure,¹⁵² resulting in tracheal dilation, tracheomalacia, and injury to the adjacent great vessels and esophagus.

Tracheal Rupture and Tracheal Laceration

Tracheal laceration and tracheal rupture are rare complications of TLI during the period when the ETT is in place.

Tracheal Dilation

Dilation of the trachea, a sign of severe tracheal wall damage,¹⁵² may appear even with soft-cuff ETTs during TLI or after extubation (See the section [Late Complications After Extubation/Tracheal/Tracheal Dilation](#)),¹⁵³ although this complication is mainly related to tracheotomy.¹⁵⁴ When tracheotomy is performed following long-term TLI, it may be difficult to determine whether the cuffed ETT or the cuffed tracheostomy tube was primarily responsible for the injury.¹⁵⁵

Tracheomalacia

Although tracheomalacia has been reported in patients with tracheotomy following TLI,¹⁵⁶ tracheomalacia after TLI alone is very rare when ETTs with high-volume, low-pressure cuffs are used.

Tracheoesophageal Fistula

Cuff pressure–induced necrosis and erosion of the posterior membranous portion of the trachea and anterior wall of the esophagus may result in tracheoesophageal fistula.^{132,157,158} This rare complication of artificial airways has been mainly associated with tracheotomy,^{132,159,160} but TLI, even with soft-cuff ETTs, is also a risk factor.¹⁶¹ The concurrent use of a nasogastric tube could conceivably enhance the risk. Tracheoesophageal fistula may be silent or present with the appearance of gastric contents in tracheal secretions or with massive gastric distension.¹⁶² After cuff deflation or extubation, it may present with cough following ingestion of food.¹⁶¹

Tracheoarterial Fistula

Tracheoarterial fistula is a very rare complication associated with long-term tracheotomy. This usually fatal event has been attributed to high cuff inflation pressure leading to tracheal necrosis and erosion of the wall of the innominate artery. Only a few cases of tracheoarterial fistula during prolonged TLI have been reported.^{163–165} In one of these cases, erosion into the innominate artery occurred at the level of the tip of the ETT, leading to fatal exsanguination on the fortieth day of TLI.¹⁶⁵ There is one report of tracheoinnominate artery fistula complicating short-term anesthetic orotracheal intubation.¹⁶⁶

Epithelial Damage

Contact of the tracheal surface with an inflated cuff, ETT tip, or suction catheter results in loss of columnar tracheal epithelium cells and distortion of their cilia.^{167–169} Regeneration of the ciliated surface and restoration of ciliary structure and function occurs within 2 weeks after extubation^{167,168}

Squamous Metaplasia of the Tracheal Epithelium

Squamous metaplasia of the tracheal mucosa has been observed in dog models of intubation and in humans^{147,170,171} and is attributed to abrasion of the mucosa by the inflated cuff or suction catheter. It may involve both surface and glandular epithelium adjacent to necrotic tracheal ulcers following prolonged TLI.¹⁷²

Reduction in Mucociliary Clearance

Tracheal injury at the cuff site depresses mucociliary clearance in both experimental animals and humans. Cuff inflation,¹⁶⁸ squamous metaplasia,¹⁷³ and loss of cilia¹⁶⁹ all contribute to impaired mucociliary transport during TLI.

Airway Colonization with Bacteria

Colonization of the trachea with bacteria occurs in nearly all patients during TLI,^{174–176} mainly from aspiration of secretions above the inflated ETT cuff and from infected ETT biofilm.¹⁷⁷ Gram-negative bacteria, particularly *Pseudomonas* species, have a particular tropism for the tracheobronchial mucosa.

Bacterial colonization of the trachea during TLI is a risk factor for ventilator-associated pneumonia (VAP; see [Chapter 46](#)).^{177,178} Ventilator-associated tracheobronchitis in intubated patients (fever, purulent lower airway secretions, positive sputum cultures, and absence of new or progressive lung infiltrates) is also increasingly recognized as an important clinical infection in the ICU.^{179,180}

Pseudomembranous Tracheitis

Pseudomembranous tracheitis, a rare and little known complication of TLI that has been attributed to ischemic cuff-site injury, was first reported in 1969.¹³¹ Lins et al reported five cases and reviewed a total of twenty-four cases.¹⁸¹ Tracheal epithelial injury from aspiration of gastric contents during TLI has been also cited as a cause of tracheal pseudomembrane formation.¹⁸² Harbison et al reported a patient who presented with acute stridor from a fibrinous tracheal membrane 3 days after TLI lasting 48 hours.¹⁸³ Removal of the membrane by rigid bronchoscopy was curative. Deslee et al described one fatal case of fibrinous tracheal pseudomembrane from TLI.¹⁸⁴

Miscellaneous

Tracheobronchitis, protracted coughing, and consequences of ETT suctioning,^{89,185} such as chest pain, hypoxemia, and cardiac arrhythmias, are sometimes observed.

Bronchial

Main-stem bronchus intubation (see [Fig. 39-3](#)) is mainly a complication during ETT placement, but it may also occur after intubation. In a retrospective series of 278 adult ICU patients with TLI, Kollef et al found that seven (2.5%) had inadvertent endobronchial ETT placement.¹⁸⁶

Pulmonary

Pulmonary complications during TLI include aspiration, pneumonia, retained secretions, and atelectasis.

Aspiration

Aspiration of oropharyngeal secretions or gastric contents can be a serious complication while the ETT is in place.^{35,177,187-189} Leakage of contaminated subglottic secretions around the ETT cuff into the lower airway increases the risk of VAP.¹⁷⁷ Estimates of the frequency of clinically significant aspiration during TLI range as high as 20%.¹⁸⁸ Aspiration of gastric contents during TLI may be less common than aspiration of oral and pharyngeal secretions, but data are sparse.

Pneumonia

VAP (see [Chapter 46](#)) is a common and serious threat to patients with artificial airways who require mechanical ventilation.^{35,177,178,190} VAP occurs in as many as 28% of patients during the course of mechanical ventilation.¹⁹⁰ A large, retrospective, matched cohort study showed that VAP developed in 9.3% of adult patients receiving mechanical ventilation for more than 24 hours.¹⁹¹ The mortality rate of VAP ranges from 24% to 50%.¹⁹⁰ Mechanisms of pneumonia related to TLI are discussed (see the section [Pathogenesis of Complications of Translaryngeal Intubation/Complications While the Endotracheal Tube Is in Place/Clinical Influences in Pulmonary Complications/Pneumonia and Tracheobronchitis](#)).

Retained Secretions

TLI may promote retention of bronchopulmonary secretions in the lung because of impaired cough performance.¹⁹²

Atelectasis

Atelectasis may complicate TLI as a result of retained bronchopulmonary secretions, migration of the ETT into a main-stem bronchus, and other factors operative during mechanical ventilation.

Miscellaneous

General

Pain, Discomfort, and Psychological Impact

Certainly all who have cared for intubated patients recognize that anxiety, depression, feelings of isolation and withdrawal, and frustration with impaired communication are major psychological detriments related to TLI. The immediate and long-term psychological impact of TLI, however, has been the subject of only a few investigations. The psychological health of intubated patients is influenced not only by TLI, but also by noise, light, sleep deprivation, stresses of underlying illness, and other adverse influences in the ICU.¹⁹³ In a study that was not entirely limited to TLI patients, Swaiss and Badran recorded the following distressing complaints in interviews of adult patients 1 day after discharge from the ICU: anxiety (68%), discomfort from the ETT (60%), fear (54%),

pain (52%), discomfort from the nasogastric tube (48%), difficulty in communicating (33%), dreams and hallucinations (31%), discomfort from physiotherapy (24%), noise (15%), insomnia (13%), and thirst (10%).¹⁹⁴

Unplanned Extubation

“Unplanned extubation” is a broad categorical term that includes self-extubations and other extubations that are variously called “accidental,” “deliberate,” and “inadvertent” in the literature.^{19,64,107,195–197} The overall frequency of unplanned extubations in critically ill adults is 8.6%.¹⁹⁷ In a prospective epidemiologic study of more than 5000 critically ill adults with TLI, self-extubation was the most common accident related to TLI.¹⁹⁸ Unplanned extubation is more likely to occur with oral than with nasal TLI.^{199,200} In a large case series, Chevron et al reported that unplanned extubation was usually deliberate (87% of patients) and associated with insufficient sedation.²⁰⁰ Unplanned extubation may occur from forces applied to the ETT by ventilator tubing, bedrails, patient motion, bucking, coughing, and forcing the tube forward with the tongue.

Coppolo and May reported a 12-month experience with self-extubation in adult ICU patients and noted self-extubation in 11%, cardiorespiratory sequelae of self-extubation in 31%, and need for reintubation in 31% of cases.¹⁹⁵ Others have documented reintubation rates following self-extubation as high as 74%.²⁰¹ Not surprisingly, the need for reintubation is much lower when self-extubation occurs during weaning from mechanical ventilation than during the period of full ventilator support.²⁰² Besides the need for reintubation, sequelae of unplanned extubation include longer ICU and hospital stays,^{199,203,204} prolongation of mechanical ventilation,²⁰³ and complications of TLI.²⁰⁴

Mechanical Complications

Various mechanical problems with ETTs occur in approximately 6% of intubations (Table 39-5),¹⁹ and they may place patients at risk for serious complications and need for reintubation. Beckmann and Gillies examined a large database of voluntary incident reports from ICUs in Australia and New Zealand to determine the frequency of reintubations not related to accidental or self-extubation.²⁰⁵ Among 143 such incidents, the reasons for reintubation included the following mechanical complications: tube malposition (twenty-five [17%]), tube securing or taping problems (twenty-four [16%]), pilot tube or cuff problem (twenty-three [16%]), and blocked or kinked airway (twenty [14%]), in addition to failed extubation (nineteen [14%]), inadequate preparation for extubation (eight [6%]), and other events (twenty-four [17%]).²⁰⁵

Table 39-5: Mechanical Complications While the Endotracheal Tube Is in Place

Tube dislocation
Endobronchial intubation
Self-extubation
Tube obstruction
From tube and cuff problems
Kinking of nasotracheal tube
Overinflation of the cuff
Herniation of the cuff over the distal end of the tube
From external compression
Biting
Cuff overinflation
From impaction of the tip of the tube on the tracheal wall
From internal obstruction
Retained secretions
Blood and blood clots
Foreign body
Tumor
Anesthetic jelly
Nasal turbinate
Nasal polyp
Disconnection from the ventilator
Difficulty passing suction catheters
Cuff problems
Leak
Rupture
Inability to seal the airway in spite of high cuff inflation pressures
Malfunction of the one-way inflation valve
Severance of the inflation tube or pilot balloon

Dislocation or misplacement of an ETT may result in endobronchial intubation (see Fig. 39-3), carinal irritation by the ETT tip, high placement at the level of the upper trachea or hypopharynx, and self-extubation. Potential complications of high tube placement include injury to the larynx and adjacent structures including the recurrent laryngeal nerves, inability to seal the airway, leading to aspiration and inadequate ventilation, and subglottic stenosis. Kollef et al reported that twenty-two (8%) of 278 adult ICU patients had at least one significant ETT dislocation, with complications such as anoxic encephalopathy, hypoxemia, gastric aspiration, atelectasis, and pneumothorax.¹⁸⁶ An earlier prospective study revealed ETT malposition in five (8.2%) of sixty-one patients with prolonged TLI.³⁵

If an ETT is positioned too low in the trachea, flexion of the neck may allow it to migrate into a main-stem bronchus, and conversely, if an ETT is positioned too high in the trachea, extension of the neck may allow it to migrate into the pharynx, potentially leading to unplanned extubation.²⁰⁶ Conrardy et al showed that flexion of the neck moved the tip of the ETT an average of 1.9 cm toward the carina (range: 0 to 3.1 cm), and extension of the neck moved the ETT a similar distance away from the carina (range: 0.2 to 5.2 cm).²⁰⁶ The route of intubation (nasal vs. oral) and cuff inflation or deflation did not affect tube movement.

The ETT may become partially or completely obstructed during placement or while it is in place by patient biting, retained secretions, blood or blood clots, foreign bodies, tumor, or other material. Obstruction from tube kinking or external compression by cuff overinflation is very rare. The tip of the ETT may impact on the tracheal mucosa leading to tube obstruction if the tube lacks a Murphy eye.²⁰⁷

ETT cuff problems during TLI include leak, rupture, and inability to seal the airway. Rashkin and Davis noted leaky cuffs during TLI in fifteen (25%) of their sixty-one patients.³⁵ Cuff rupture is now rarely observed. Inability to seal the airway by inflating or overinflating the cuff was the most common early clinical problem following TLI in one series.¹⁹ Vyas et al checked cuff pressures in thirty-two patients intubated with ETTs with high-volume, low-pressure cuffs and found that 62% had elevated cuff pressures, some as high as 100 cm H₂O.²⁰⁸ They performed a telephone survey that revealed that cuff pressure was checked regularly in only 17% of ICUs.

A wide variety of other mechanical problems have been encountered while the ETT is in place, including malfunction of the one-way inflation valve, inadvertent severance of the inflation tube or pilot balloon, disconnection from the ventilator, and difficulty passing a suction catheter through the ETT.

Malnutrition

TLI precludes effective swallowing and necessitates feeding by way of enteric tubes or intravenous lines. [Chapter 41](#) discusses malnutrition in patients with TLI.

Esophageal

Tracheoesophageal fistula may develop during TLI (see the section [Complications While the Endotracheal Tube Is in Place/Tracheal/Tracheoesophageal Fistula](#)).

Gastric

Gastric distension may occur if a tracheoesophageal fistula develops during TLI.¹⁶²

Musculoskeletal and Neurologic

Musculoskeletal and neurologic complications directly related to TLI while the ETT is in place have not been reported.

Physiologic

Compared with a normal upper airway, an ETT increases the resistance to airflow. Gal and Suratt demonstrated that TLI in healthy men increased total airways resistance from 0.99 cm H₂O/L/s to 2.25 cm H₂O/L/s, an increase of 178%.^{209,210} Approximately half of this increase was attributed to the ETT itself, and the remainder to reflex bronchoconstriction.^{209,210} TLI increases lower airway resistance slightly in normals without reactive airways disease by stimulation of irritant receptors, provoking bronchoconstriction,²¹¹ and prophylactic treatment with bronchodilators has a protective effect.²¹²

Airflow resistance during TLI increases as the length of the ETT increases or its diameter decreases.²¹³ Because they increase airflow resistance, ETTs increase the patient's work of breathing.²¹⁴ The work of breathing increases as the patient's minute ventilation increases and as the diameter of the ETT decreases.^{214–216} At minute ventilation rates at or below 8 L/min, an ETT has little impact on the work of breathing.²¹⁵ Although the extra work of breathing imposed by an ETT is not clinically important in most patients, it may be clinically significant in patients with poor ventilatory function.

Complications during and after Extubation

Complications during Extubation

Oral, nasal, pharyngeal, and laryngeal complications that developed while the ETT was in place may, of course, first be detected at extubation. [Table 39-6](#) lists complications of TLI that Stauffer et al observed immediately after extubation in the ICU.¹⁹ Hoarseness and sore throat are the most common symptoms at extubation. New complications occurring in the brief period of extubation are transient and uncommonly reported. They include discomfort, pain, and anxiety.¹⁹³ Significant increases in heart rate, blood pressure, and plasma catecholamine levels occur at extubation.⁹⁴ Nasal bleeding and dental injuries may occur during extubation, but such events are rarely reported.

Table 39-6: Clinical Findings at Extubation in a Series of 81 Patients

	No.	Percent (%)
History (obtained in sixty-nine patients)		
Symptoms related to TLI	59	86
No symptoms related to TLI	10	14
Specific symptoms		
Hoarseness	49	71
Sore throat	29	42
New cough	18	26
New sputum production	15	22
Hemoptysis	7	10
Other upper airway complaints	9	13
Physical examination (obtained in seventy-eight patients)		
Abnormal, related to TLI	42	54
Normal	36	46
Specific abnormalities		
Auscultatory abnormalities over the trachea	20	26
Lip ulcer or cellulitis	12	15
Pharyngeal bleeding or ulceration	7	9
Ulceration of the palate	6	8
Nasal bleeding	6	8
Oral mucous membrane bleeding, ulceration, or inflammation	5	6
Stridor	4	5
Nasal ulceration	2	3
Tooth avulsion	2	3
Tongue injury	2	3

Source: Reprinted from *Am J Med*, Vol. 70, Stauffer JL, Olson DE, Petty TL.

Complications and consequences of endotracheal intubation and tracheotomy: a prospective study of 150 critically ill adults, pages 65–76. Copyright © 1981, with permission from Elsevier.

Partial laryngeal obstruction at extubation is most often related to laryngeal edema (see the section [Early Complications After Extubation/Laryngeal/Edema](#)). Laryngeal injury and other complications during self-extubation of an ETT with an inflated cuff have not been adequately studied. Surprisingly, few patients appear to develop laryngeal complications following self-extubation. Pulmonary aspiration may occur if extubation and suctioning techniques are faulty. The

requirement for reintubation soon after extubation is usually caused by laryngeal edema or respiratory failure. Difficult extubation should be anticipated in patients whose intubation was difficult.²¹⁷

Early Complications after Extubation

This section discusses those complications of TLI that appear during roughly the first week after extubation. This period is dominated by symptoms of pharyngeal and laryngeal injury from TLI, including hoarseness, dysphonia, sore throat, cough, dysphagia, and odynophagia.

Nasal, Paranasal, and Oral

There are no specific new complications that develop in this time interval, but complications that began earlier may of course be observed.

Pharyngeal

Postextubation sore throat has been reported in up to 42%¹⁹ of ICU patients, similar to the frequency observed in patients who required TLI for general anesthesia.^{218,219} Persistent and severe dysphagia after extubation following prolonged TLI should alert the clinician to the possibility of active infection in the posterior larynx.

There may be a temporary loss of the gag reflex after extubation. This was observed in 24% of patients who had direct laryngoscopy within 24 hours of extubation, supporting the notion that some patients have a sensory denervation of the pharynx and larynx from TLI.⁵² Loss of upper airway reflexes increases the risk of aspiration.⁵² We found pharyngeal bleeding or ulceration in 9% of patients at extubation (see Table 39-6),¹⁹ although Colice et al did not report this complication after prolonged TLI.⁵²

Laryngeal

Prospective studies using fiber-optic laryngoscopy within 24 hours of extubation in the ICU setting have revealed laryngeal injury in 73% to 94% of patients.^{52,220,221} Colice et al performed direct fiber-optic laryngoscopy within 24 hours of extubation in eighty-two patients who had been intubated for 4 or more days (mean: 9.7 ± 0.6 days).⁵² Seventy-seven (94%) of their patients had laryngeal injury (42% mild, 29% moderate, and 23% severe). Nearly all patients had posterior true vocal cord ulceration. Laryngeal edema was present in the patients with moderate to severe damage. Two patients had such large ulcers that the vocal cords could not be adducted, permitting free aspiration into the trachea.⁵² Most patients who displayed initial laryngeal damage had complete resolution of laryngeal findings on laryngoscopy within 4 weeks after extubation.⁵² Kastanos et al reported the following early laryngeal lesions in nineteen patients after extubation: true vocal cord granulomas in eight (42%), true vocal cord ulcerations in seven (37%), true vocal cord paresis in one (5%), and subglottic edema in one (5%).¹²⁸ Thomas et al performed fiber-optic endoscopy through the ETT just before extubation and found visible laryngeal pathology in 131 (88%) of 150 patients.²²² Table 39-7 summarizes the important laryngeal complications after TLI.

Table 39-7: Important Early and Late Laryngeal Complications after Extubation

Early
Hoarseness
Edema
Stridor
Muscle dysfunction
Vocal cord paresis and paralysis
Vocal cord hematoma
Posterior glottic infection and cricoid cartilage abscess
Late
Glottic and subglottic stenosis
Granulomas
Synechiae

Hoarseness

In fifty-four adult male patients followed closely by Colice et al after TLI, thirty (56%) had significant hoarseness.⁵² Holdgaard et al observed hoarseness in 42% of ICU patients following nasotracheal intubation.¹⁰⁵ Hoarseness following TLI usually resolves within a few days in most patients as laryngeal edema and inflammation subside and normal vocal cord adduction is restored.¹²⁵ In some cases, transient postextubation hoarseness requires 7 to 10 days to

resolve. Hoarseness persisting beyond 10 to 14 days suggests serious laryngeal injury such as granuloma formation, vocal cord paresis or paralysis, or cricoarytenoid joint dysfunction, and requires the attention of an otolaryngologist–head and neck surgeon.

Edema

Postextubation laryngeal edema (see Fig. 39-11), an important complication of TLI, may occur at the supraglottic, glottic, or subglottic levels.⁸ It presents with postextubation stridor and dyspnea and commonly precipitates the need for reintubation. Laryngeal edema is observed immediately after extubation in approximately 40% of patients after anesthetic intubation and in a roughly a third²²³ to half of critical care patients.⁵²

Laryngospasm

Postextubation laryngospasm in the recovery from general anesthesia is a well-known phenomenon. Although postextubation stridor in the ICU is sometimes attributed to laryngospasm, this complication after extubation of ICU patients is not well documented. Colice et al reported true laryngospasm in only one of eighty-two ICU patients within 24 hours of extubation.⁵² True laryngospasm following extubation of the critically ill patient should be distinguished from laryngeal edema and other more common causes of postextubation stridor.

Stridor

A literature review by Wittekamp et al reported postextubation stridor in up to 30.2% of critically ill adults.¹⁴⁴ Many studies have investigated postextubation stridor.^{19,52,224–226} Stridor occurring after extubation is usually inspiratory, indicating variable extrathoracic airway obstruction. Mackenzie et al observed severe stridor after prolonged TLI in 0.1% to 0.6% of patients.²²⁴ This complication was found to be caused by vocal cord edema or subglottic stenosis, not laryngospasm. Symptoms of stridor appeared from 5 minutes to 4 hours after extubation.²²⁴ Repeated intubations or tracheotomy were required in the majority of patients. Postextubation stridor should always indicate the possibility of subglottic stenosis at the level of the cricoid cartilage, a level that is particularly vulnerable to airway narrowing in children.

Muscle Dysfunction

Laryngeal incompetence is common after extubation and has been attributed to a sensory impairment or mechanical dysfunction of the larynx.²²⁷ The frequency of aspiration secondary to laryngeal incompetence in postoperative patients ranges from 22% to 35%.^{227,228} Inability of the vocal cords to adduct increases the risk of aspiration.^{52,125}

Vocal Cord Paresis and Paralysis

Unilateral vocal cord paresis or paralysis is a serious but uncommon complication of TLI.^{47,125,229,230} It presents in the early postextubation period as persistent hoarseness. Santos et al reported vocal cord immobility in sixteen (20%) of ninety-seven patients after extubation, and in half of these cases, a delayed presentation over a period of 1 to 10 weeks following extubation was observed.¹³⁹ Others have reported vocal cord paralysis in less than 1% of cases.⁴⁷ True bilateral vocal cord paralysis after TLI, which presents as severe stridor, is very rare.

Whited evaluated sixteen patients with vocal cord paresis or paralysis after TLI of 5 days or longer.¹²⁵ Common features in these patients were symmetry of vocal cord paresis or paralysis, edema and erythema of the arytenoids and posterior commissure, median or paramedian vocal cord positioning, late return of vocal cord abduction, a tendency for aspiration to occur, and spontaneous recovery in nearly all patients within 4 weeks.¹²⁵ Postextubation vocal cord paresis or paralysis may result from traumatic dislocation of the arytenoids,^{231,232} cricoarytenoid joint arthritis, cuff inflation immediately below the vocal cords resulting in compression of the anterior branch of the recurrent laryngeal nerves,²³³ or the mechanical effects of laryngeal edema and inflammation.¹⁰⁸ Bilateral vocal cord paralysis from intubation injury may present with hoarseness and recurrent aspiration²³³ or with upper airway obstruction requiring immediate tracheotomy.

Vocal Cord Hematoma

Hematomas of the vocal cords that occur during placement of an ETT apparently resolve by the time of extubation. They were not seen after extubation in eighty-two patients studied by Colice et al.⁵²

Posterior Glottic Infection and Cricoid Cartilage Abscess

Severe odynophagia or odynophonia after extubation should suggest the possibility of posterior glottic infection or cricoid cartilage abscess.¹⁰⁸ Posterior glottic infection may result in destruction of laryngeal cartilage and predispose to posterior glottic stenosis.¹⁰⁸

Tracheal

In a prospective study of nineteen patients, Kastanos et al reported finding tracheal granulomas in six patients (31%) and cuff-site tracheitis in three (16%).¹²⁸ Tracheal pseudomembranes leading to airway obstruction in the first few days after extubation have been the subject of individual case reports.^{181,182}

Bronchial

Specific bronchial complications of TLI in this time period have not been reported.

Pulmonary

Pulmonary aspiration may occur with laryngeal incompetence after extubation.^{52,125,227} Pulmonary aspiration was observed after extubation in two (2.4%) of eighty-two patients after prolonged TLI in the series of Colice et al.⁵² This finding was attributed to vocal cord dysfunction and temporary loss of sensorimotor function of the larynx and pharynx.⁵² Burgess et al observed aspiration of swallowed contrast material in 33% of surgical patients immediately after extubation.²²⁷ Endoscopic evaluation of swallowing after prolonged TLI has revealed aspiration in 14%²³⁴ to 45%²³⁵ of patients.

Postextubation negative pressure pulmonary edema secondary to laryngeal or tracheal obstruction following TLI and prolonged forceful inspiration has been described in small case series or individual case reports.^{236,237}

Miscellaneous

There are no significant miscellaneous complications of TLI in the early postextubation period.

Late Complications after Extubation

Late complications of TLI, those appearing in the weeks to months after extubation, are usually the result of abnormal airway mucosal healing with fibrosis or granuloma formation. Laryngeal stenosis is the most serious of these events (see discussion below).

Nasal and Paranasal

Nasal stricture and nasal septal perforation following prolonged nasotracheal intubation are rare.

Oral

The only late oral complications of TLI are those related to initial dental or oral trauma.

Pharyngeal

In some patients, dysphagia may persist for several weeks. Residual problems related to pharyngeal perforation during ETT placement may persist late after extubation.

Laryngeal

Fortunately, late laryngeal complications of TLI (see Table 39-7) are uncommon because of the normal healing of laryngeal ulcers and other laryngeal injuries discussed above. Healing of laryngeal mucosal injury after extubation is usually complete within 8 to 12 weeks.¹³⁸ Nevertheless, abnormal healing may result in serious late laryngeal complications of TLI. Hoarseness that fails to improve for many weeks after extubation suggests abnormal healing with structural deformity of the larynx.¹³⁰

Laryngeal Stenosis

Laryngeal stenosis following TLI may be supraglottic, glottic, or subglottic in location. Supraglottic stenosis is unusual. Glottic or subglottic stenosis is a serious but fortunately uncommon late complication of TLI. Although rarely severe, it represents the most notorious complication of TLI and has received extensive attention in the literature.^{52,108,125–127,130,238–242} TLI is the most common cause of posterior glottic laryngeal stenosis.²⁴²

Most prospective studies of TLI document a very low rate of laryngeal stenosis.^{19,128,130,138,243} The reported frequency of this complication ranges from 0% to 12%. When the results of seven prospective studies are combined, the overall incidence of laryngeal stenosis is 1.3% to 2.9% (Table 39-8), depending on whether the disproportionately higher number of cases in the Whited analysis is included. Colice found no cases of laryngeal stenosis in fifty-four prospectively studied survivors of prolonged TLI.¹³⁸ Elliott et al found that three (10%) of thirty survivors of the acute respiratory distress syndrome had laryngeal stenosis.²⁴⁰ Laryngeal stenosis following TLI for general anesthesia is very rare.

Table 39-8: Rates of Laryngeal Stenosis after Prolonged Translaryngeal Intubation in Prospective Studies

Year	Reference	Frequency of Laryngeal Stenosis
1969	Lindholm ¹³⁰	1/206 (0.5%)
1981	Stauffer et al ¹⁹	2/27 (7.4%)
1982	Pecora and Seinige ²⁴³	0/21 (0%)
1983	Kastanos et al ¹²⁸	2/19 (10.5%)
1983	Whited ¹²⁶	12/200 (6.0%)
1989	Colice ¹³⁸	0/54 (0%)
1994	Santos et al ¹³⁹	0/62 (0%)
	Total	17/589 (2.9%) ^a

^aWith the exception of the study by Whited, the rate of laryngeal stenosis is five of 389 (1.3%).

Whited found posterior commissure fibrosis with stenosis in two (12.5%) of sixteen patients with TLI of 5 days or longer who were referred for evaluation of vocal cord paralysis or paresis.¹²⁵ Later, he reported frequencies of laryngeal stenosis of 6% in 200 patients with prolonged TLI¹²⁶ and 14% in fifty patients intubated for 11 days or longer.¹²⁷ This rate of laryngeal stenosis is higher than the rates in other prospective studies (see Table 39-8). In Whited's study, duration of TLI was the only independent variable assessed, and other potential risk factors for laryngeal stenosis were not evaluated.^{126,127}

Lindholm noted only one case of laryngeal stenosis (subglottic) in 206 adult patients after TLI,¹³⁰ but patients in his series had a relatively short duration of intubation (mean: 32 hours). Stauffer et al found that two (7%) of twenty-seven patients evaluated for airway stenosis with tomograms after extubation had subglottic stenosis, defined as a 10% or greater reduction in transverse diameter of the trachea at the cricoid level.¹⁹ Subglottic stenosis in adults following prolonged TLI appears to be much less common than in neonates and children, in whom this complication occurs with a frequency as high as 8%.²⁴⁴

Laryngeal Granuloma Formation

Granuloma formation is an important late complication of TLI, typically presenting as persistent hoarseness. Granulomas occur primarily in the larynx,^{19,130,139,245} but rarely may occur in the trachea. Laryngeal granulomas (see Figs. 39-9B and C and 39-12) are usually a few millimeters in diameter and rarely grow large enough to compromise the airway lumen. They may be sessile or pedunculated. As a manifestation of the healing process, albeit abnormal, granulomas may not appear until weeks after extubation. Laryngeal granulomas usually appear at the edges of posterior glottic ulcers, particularly those on the vocal processes and arytenoids. They usually resolve spontaneously, but surgical excision is sometimes required because of persistent hoarseness or airway compromise.¹³⁸

The reported frequency of laryngeal granulomas after TLI is highly variable. After anesthetic intubation, they are extremely rare.²⁴⁵ Colice et al found laryngeal granulomas in four (7%) of fifty-four patients carefully examined after TLI.⁵² In that series, all patients with persistent hoarseness after TLI had laryngeal granulomas. Lindholm found postintubation granulomas in approximately one-third of adults after TLI, but not in any infants or children.¹³⁰ In the Lindholm series, only seven (2.6%) of 265 patients followed after extubation required surgical excision of laryngeal granulomas. The report of Santos et al of ninety-seven patients with prolonged TLI described laryngeal granulomas in 44%, and in 57% of these cases, the granulomas developed an average of 4 weeks after extubation.¹³⁹

Synechia Formation

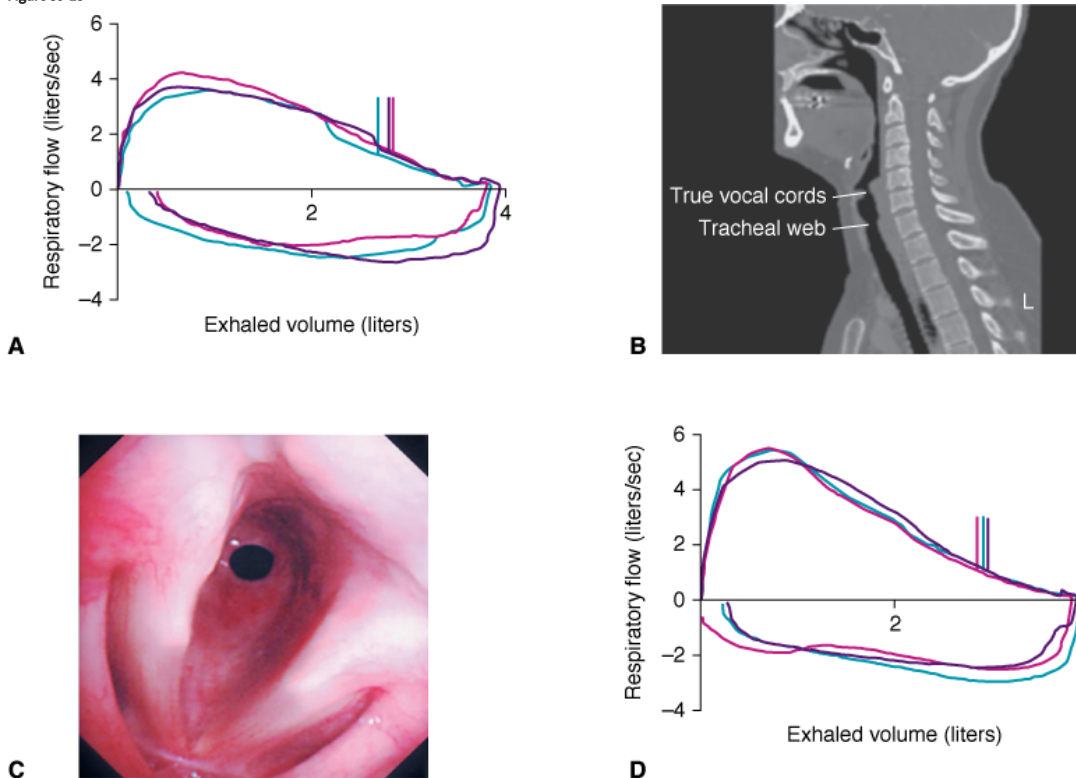
Synechia (membrane or web) formation at the glottis occurs in less than 1% of TLIs.^{130,246} Synechia formation may “weld” the vocal cords together, leading to aphonia and airway obstruction.⁸

Tracheal

Tracheal Stenosis

Tracheal stenosis is an uncommon complication of TLI alone. [Figure 39-13](#) illustrates endoscopic and clinical features of tracheal stenosis in a patient with dyspnea following TLI.²⁴⁷ Stenosis of the trachea in adults surviving prolonged periods of mechanical ventilation is usually attributed to tracheotomy rather than TLI.²⁴⁸ Severe tracheal stenosis following TLI alone is fortunately rare.²⁴⁹ Santos et al prospectively observed no cases of tracheal stenosis in sixty-two adults surviving long-term TLI.¹³⁹ A retrospective review of laryngotracheal stenosis following TLI, tracheotomy, or both in 315 neurologic patients reported by Richard et al found tracheal stenosis in only one patient in the group of 172 patients (0.6%) with TLI only (mean duration: 17 days).²⁴¹ In contrast, twenty-five (19%) of 131 patients with TLI followed by tracheotomy had tracheal stenosis. Tracheal stenosis following tracheotomy usually occurs at the site of the tracheal stoma,²⁵⁰ whereas this lesion is observed at the site of the inflated cuff when it follows TLI alone. Spiral computed tomography with multiplanar reconstruction and virtual bronchoscopy are highly accurate in diagnosing tracheal stenosis.²⁵¹

Figure 39-13



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Tracheal stenosis is an important but uncommon complication of endotracheal intubation. Panel C illustrates subglottic tracheal stenosis in a 45-year-old woman with dyspnea who had required prolonged endotracheal intubation 8 years earlier. Computed tomography demonstrated tracheal narrowing 2.8 cm below the vocal cords (B), and a flow-volume loop revealed attenuated expiratory/inspiratory flow rates (A). Marked improvement in symptoms and stridor and improvement in flow rates (D) were reported after resection of the stenotic segment and tracheal dilation. (Reproduced, with permission, from Datta A, Cale A. Postoperative tracheal stenosis. *N Engl J Med*. 2010;362:e5, January 14, 2010. Copyright © Massachusetts Medical Society.)

Tracheal Dilation

Persistent tracheal dilation is caused by cartilaginous injury at the cuff site.¹⁵²⁻¹⁵⁴ This complication was reported in 2% to 5% of patients following use of hard-cuff tracheostomy tubes,¹⁵³ and it is rarely seen following TLI with soft-cuff ETTs.

Tracheal Granuloma Formation

Tracheal granulomas occur mainly at the site of tracheal cuff-site ulcers. These are extremely rare in the era of soft-cuff ETTs.

Bronchial

There are no late bronchial complications of TLI.

Pulmonary

There are no late primary pulmonary complications of TLI.

Miscellaneous

There are no important late miscellaneous complications of TLI other than residual problems related to tracheoesophageal fistula.

Pathogenesis of Complications of Translaryngeal Intubation

This section reviews the causes and mechanisms of selected complications of TLI. Most of the investigation of mechanisms of complications of TLI has focused on laryngeal injury and tracheal cuff-site injury.

Complications during Endotracheal Tube Placement

Complications during placement of an ETT often occur as a result of suboptimal technique stemming from poor judgment or inexperience. Failure to prepare the patient for intubation and equipment malfunction may also lead to complications. For example, during nasal intubation attempts, failure to prepare the nasal mucosa with a topical anesthetic or vasoconstrictor or failure to select the more patent side of the nose for intubation may result in nasal mucosal laceration or turbinate injury.

It is generally taught that complications of intubation are more likely to occur in emergency situations than in elective, controlled intubations, but there are few data to establish this with certainty. Wang et al found that failure of prehospital intubation attempts was significantly associated with a number of covariates by multivariate logistic regression, including the presence of clenched jaw/trismus, inability to pass the ETT past the vocal cords, intact gag reflex, and increased body weight.²⁵² Adnet et al found significant associations between the number of intubation attempts and the rate of mechanical and general complications of emergency prehospital TLI,²⁹ but Schwartz et al did not find this association in critical care intubation.⁷⁰ Supervision of TLI in teaching hospitals by senior physicians is associated with a lower rate of complications.²⁰

Incorrect use of the laryngoscope may cause dental injury or pharyngeal laceration. Blind jamming of the ETT into the hypopharynx may cause serious soft-tissue injury. Failure to oxygenate and ventilate the patient before intubation may result in hypoxic brain injury. Deep insertion of the ETT far beyond the vocal cords may result in right main-stem bronchus intubation. These are only a few examples of how faulty intubation technique, inexperience, and poor judgment may lead to intubation complications.

Complications While the Endotracheal Tube Is in Place

Complications of TLI during this time interval depend upon anatomic, physiologic, ETT, and clinical influences.

Anatomic and Physiologic Influences

Nasal and Paranasal Complications

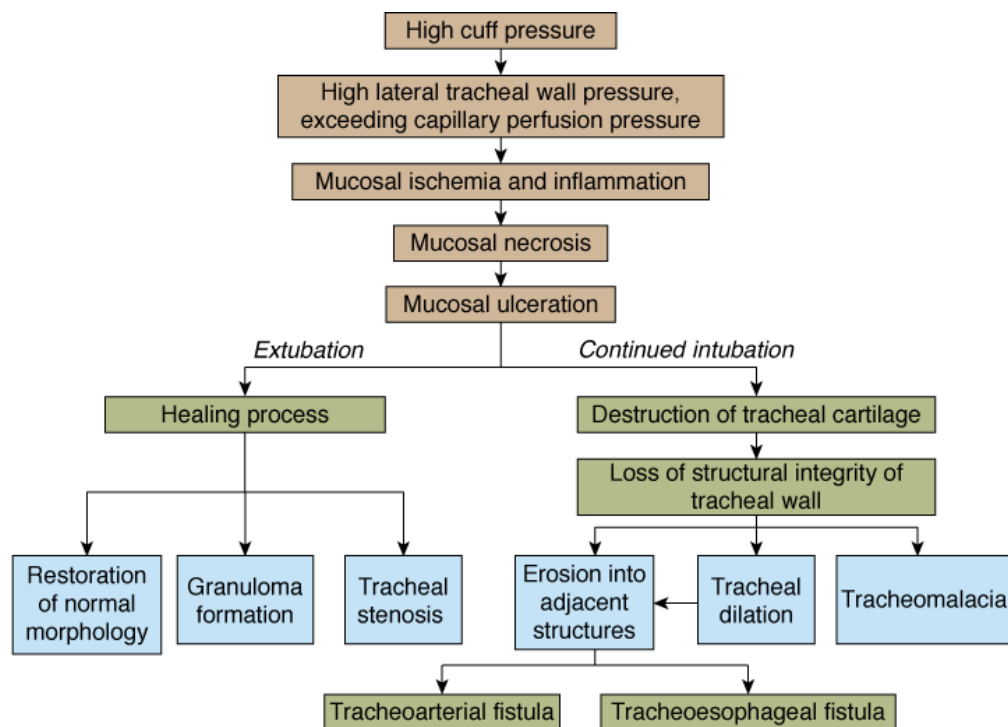
Necrosis of the nasal alae and ulceration and perforation of the nasal septum result from ischemic necrosis secondary to pressure on these structures from the nasotracheal tube. Sinus effusions and sinusitis are caused by obstruction of the sinus ostia by the adjacent nasotracheal or nasogastric tube, edema, inflammation, or mucus plugging. Otitis and middle ear effusions are related to obstruction of the eustachian tube by the same influences.

Laryngeal and Tracheal Complications

Serious laryngeal and tracheal injury during TLI is generally attributed to ischemic injury of the mucosal surface and deeper tissues by pressure exerted at vulnerable anatomic sites.²⁵³ For example, ischemia occurs when the pressure exerted on the posterolateral wall of the larynx by the ETT or on the tracheal wall by the inflated cuff exceeds the capillary perfusion pressure in these areas. Necrosis, sloughing of the mucosa, and ulceration follow. Microscopically, inflammatory cell infiltrates and bacterial invasion are seen in the devitalized mucosal tissues. Eventually, destruction of soft tissues and cartilage in the wall of the larynx and trachea occur. In severe cases, there is loss of structural integrity of the tracheal wall, producing dilation or softening. If the patient survives, the healing process begins with one of three eventual outcomes: restoration of normal morphology, development of granulation tissue, or fibrosis. In the larynx, fibrosis results in glottic or subglottic stenosis, synechia formation, or cricoarytenoid joint fixation. In the trachea, fibrosis results in segmental stenosis.

The concept of tracheal mucosal injury by an air-filled cuff was established by studies performed in the 1960s and 1970s. It is outlined in [Figure 39-14](#). Tracheal mucosal perfusion pressure is a critically important determinant of ischemic injury at the tracheal cuff site.^{254–260} In humans, the tracheal capillary perfusion pressure is estimated to be 25 mm Hg (range: 22 to 32 mm Hg) or 34 cm H₂O (range: 30 to 44 cm H₂O).^{258,259} The role of capillary perfusion pressure in the pathogenesis of tracheal cuff-site injury is discussed (see the section [Clinical Influences in Tracheal Cuff-Site Injury](#)).

Figure 39-14



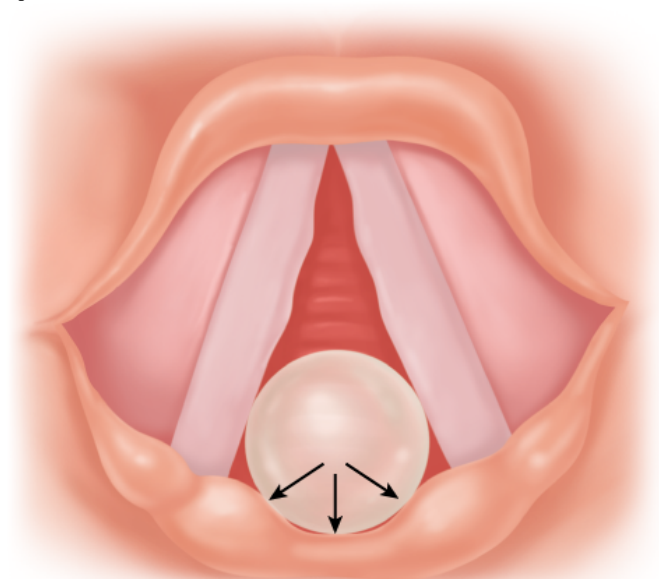
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Outline of pathogenesis of tracheal injury at the site of the air-inflated cuff.

Laryngeal and tracheal structures that are vulnerable to ETT pressure are the arytenoids, the posterior rim of the glottis, the subglottis, the cuff site, and the site of the tip of the ETT.²⁶¹ The posterior rim of the glottis is a contact point or fulcrum where the ETT abrades and applies constant pressure to the mucosa, leading to the risk of mucosal ischemia and ulceration.^{47,239} Quartararo and Bishop have described the geometric shape of the glottis as a pentagon, which of course does not resemble the circular shape of the ETT.⁴⁸ This geometric mismatch creates several points of contact where ulceration is most likely to occur (Fig. 39-15).⁴⁸ An experimental ETT was once designed to fit the natural shape of the glottis.²⁶²

Figure 39-15



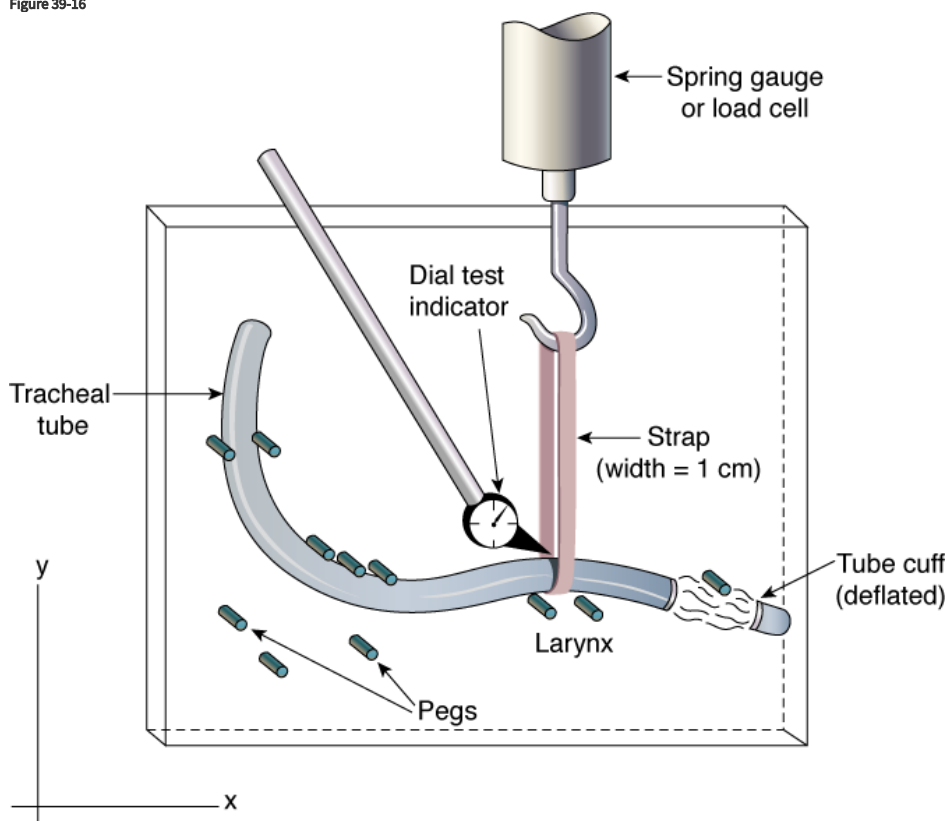
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Schematic of an endotracheal tube (ETT) in the glottis, showing the circular shape of the ETT and the pentagonal shape of the glottis. The arrows indicate the points of maximum pressure on the posterior wall of the glottic opening. (This article was published in *Semin Anesth*, Vol. 9, Quartararo C, Bishop MJ. Complications of tracheal intubation: prevention and treatment, pages 19–27, Copyright © Elsevier.)

The essential curvature of the ETT necessitates that force be directed posteriorly and laterally at the posterior glottic rim and cricoid ring.²⁵³ Furthermore, the tongue muscles force the ETT posteriorly. Glottic pressure by the ETT has been estimated in both in vitro²⁶³ (Fig. 39-16) and animal²⁶⁴ studies to range as high as 200 to 400 mm Hg. In vitro studies of a variety of ETTs indicate that deforming forces from 230 to 1000 g are applied to the posterior laryngeal wall.²⁶⁵ Pressure-induced ischemic injury of the posterior cricoid lamina and vocal processes of the arytenoids is accompanied by cricoid and arytenoid perichondritis and chondritis, which may heal with eventual fibrosis and scar contracture.²⁴²

Figure 39-16



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Method of estimating forces exerted on the posterior larynx by endotracheal tubes (ETTs). A pegboard is used to simulate the shape of the ETT in vivo. A gauge measures the force needed to barely lift the tube at the level of the larynx, thereby estimating pressure on the larynx. Modern polyvinyl chloride and silicone rubber tubes under simulated in vivo conditions exert forces on the larynx in the range of 162 to 265 g. (Modified, with permission, from Steen et al.²⁶³)

Cricoarytenoid joint injury may occur while the ETT is in place but be recognized only after extubation. Postextubation symptoms of hoarseness and odynophagia along with the findings of malposition and no movement of the arytenoids suggest cricoarytenoid joint dysfunction. Paulsen et al reported that laxity of the cricoarytenoid joint capsule and large synovial folds are predisposing factors for cricoarytenoid joint trauma from TLI, including hemarthrosis.²⁶⁶

The cricoid ring, a completely circumferential ring that encircles the lower part of the larynx, is an important contact point with an ETT. Inflammation and edema of the laryngeal mucosa at the level of the cricoid ring narrow the airway lumen, making the subglottic area vulnerable to acute stenosis. Postextubation stridor is a manifestation of this phenomenon.²²⁴ Experiments in dogs have demonstrated clearly that the posterior third of the cricoid cartilage is a major site of deep ulceration from pressure and abrasion by the ETT.²⁶³ Ischemic necrosis in this area may be followed by bacterial invasion and abscess formation.¹⁰⁸ Simultaneous use of a nasogastric tube with an ETT could conceivably increase the risk of this complication.

Whited observed that the anterior wall of the trachea is a vulnerable site for tracheal injury from an ETT because of positioning of the cuff and tip of the tube.²⁶¹ Surprisingly few complications of TLI, however, are noted in the anterior wall of the trachea in these locations in the modern era of soft-cuff tubes.

Endotracheal Tube Influences

Physical Characteristics of an Endotracheal Tube

Technical standards for modern ETTs²⁶⁷⁻²⁶⁹ establish the shape, size, and stiffness of an ETT, all of which influence TLI-induced airway injury. As previously discussed, the curved shape of an ETT invites pressure injury at the contact points with the airway, including the posterior larynx, cricoid cartilage, and

tracheal cuff and tip sites.^{130,270} The cross-sectional round shape of an ETT is also poorly suited for the shape of the glottis as noted above.^{48,271} Large ETTs are more likely than small ones to cause pressure injury to the larynx^{130,261,264} and the trachea. Large ETTs also cause more postextubation sore throat and hoarseness than do smaller tubes.²⁷² Modern ETTs vary considerably in their stiffness and thermolability (ability to become less stiff at body temperature).²⁷³

Surface Characteristics

A slimy coating or *biofilm* may accumulate on the walls of the ETT during use, promoting the adherence of oral microorganisms and placing the patient at risk for VAP.^{274,275} Use of silver-coated ETTs to suppress biofilm formation and bacterial colonization reduces the risk of VAP.²⁷⁶

Chemical Composition

The manufacture of modern ETTs follows strict guidelines set by Committee F29 on Anesthetic and Respiratory Equipment of the American Society for Testing and Materials.²⁶⁷ Modern ETTs are composed of polyvinyl chloride, which is nontoxic, but chemical additives such as stabilizers, plasticizers, fillers, and pigments may incite inflammatory reactions in tissue.²⁶⁸

Cuff Characteristics

The value of cuffs on ETTs to seal the airway became apparent in the 1930s, but in the following decade investigators observed that high cuff inflation pressures could cause ischemia and sloughing of the tracheal mucosa.²⁷⁷ Animal studies in the 1960s clarified the mechanisms of cuff-site tracheal injury,^{148,278,279} and several groups of investigators described the critical role of high cuff inflation pressures in producing ischemic necrosis of the tracheal mucosa (see the section [Clinical Influences in Tracheal Cuff-Site Injury/Cuff Pressure](#)). Others later implicated high cuff inflation pressures in mucosal inflammation, tracheal dilation, destruction of tracheal cartilage leading to tracheal stenosis and tracheoesophageal fistula, and erosion of the tracheal mucosa and submucosa with loss of submucosal glands.¹⁴⁷ By the early 1970s, investigators established clearly that inflating the ETT cuff to pressures that exceeded tracheal capillary perfusion pressure could cause mucosal ischemia. Studies of the microcirculation of the rabbit trachea by Nordin et al revealed that the mucosa overlying the cartilaginous tracheal rings would become ischemic if hard cuffs were inflated to the point where lateral tracheal wall pressure exceeded 30 mm Hg.²⁵⁷

Besides intracuff pressures, other cuff characteristics may contribute to airway injury. Overlapping folds of cuff fabric may create channels that permit migration of fluid into the lung. This aspiration process is promoted by larger tubes, larger cuff diameters, stiffer cuff fabrics, and spontaneous breathing.²⁸⁰ Bernhard et al demonstrated that modern ETTs display considerable variability in cuff characteristics such as diameter, thickness, compliance, shape, resting volume, and just-seal volume.^{273,281} These various factors, as well as the size and shape of the trachea,²⁸² may affect the pressure-volume relationship of a given ETT cuff in vivo.^{206,281}

Clinical Influences in Laryngeal Injury

Multiple factors may be important in the pathogenesis of laryngeal and tracheal cuff-site injury during TLI. Clinical investigation of the pathogenesis of airway injury from TLI has been complicated by difficulty in controlling these many factors while examining the effect of one variable on airway injury. Accordingly, the role of most of the specific clinical influences on laryngeal or tracheal injury discussed below remains unsettled.

Tube Size Relative to Laryngeal Size

The relationship of ETT size to laryngeal injury is controversial. Several prospective studies of critical care TLI have observed no relationship of laryngeal injury with ETT size.^{19,52,283} Lindholm, however, concluded that large ETT size with respect to laryngeal size is an important determinant of laryngeal injury,¹³⁰ and Santos et al observed that laryngeal erythema, ulceration, and delayed true vocal cord immobility were related to ETT size.¹³⁹

Laryngeal Abrasion

Abrasion of the laryngeal mucosa by movement at the interface between the ETT and the surface of the larynx would intuitively seem to be of great importance in the pathogenesis of laryngeal ulceration. This concept, however, has been difficult to prove. Some degree of abrasion of the laryngeal mucosa likely occurs whenever there is movement of the ETT or movement of the larynx. Breathing, swallowing, coughing, yawning, hiccupping, attempting phonation, and head and neck movements²⁰⁶ are all plausible contributors to the abrasive action of the ETT on the laryngeal surface. ETT movement with cycling of the ventilator, suctioning the airway, and nursing care interventions may further contribute to mucosal abrasion.¹²⁶

The role of laryngeal mucosal abrasion by movement of the ETT has been considered by a number of investigators.^{19,52,126,129,130,283,284} The findings of El-Naggar et al suggested that friction between the ETT and laryngeal mucosa is an important factor in the pathogenesis of laryngeal injury.¹²⁹ Colice et al

reported that the severity of laryngeal injury following extubation was significantly associated with neuromotor activity while the ETT was in place.⁵² Patients with flaccid paralysis had milder degrees of injury.⁵² Dunham and LaMonica found more laryngotracheal injury in patients with head and neck rigidity than in those with a nonrigid posture.²⁸⁴

Duration of Intubation

Since the early 1950s, clinicians and investigators have speculated, and often assumed, that duration of TLI is an important determinant of complications of TLI. This relationship obviously has an important influence on decisions about performing tracheotomy for long-term airway maintenance. It would seem obvious that a longer duration of TLI increases the risk of laryngeal (or tracheal) injury, but this relationship remains controversial because of conflicting data from a number of studies.

Early reports suggested that duration of TLI is important in the pathogenesis of laryngeal injury in critical care intubation.^{131,285–287} Donnelly observed that the severity of histopathologic injury to the larynx at autopsy was related to the duration of TLI.¹³¹ A retrospective study by Hedden et al came to the same conclusion.²⁸⁵ Later observations by Whited^{126,127} and by Supance et al²⁴⁴ suggested that longer periods of TLI increased the risk of laryngeal stenosis. Santos et al found significant relationships between duration of TLI and aspiration, laryngeal granulomas, and vocal cord immobility, but not laryngeal ulceration.¹³⁹ Two prospective studies reported a significant relationship between duration of TLI and postextubation stridor or requirement for reintubation.^{225,288}

A number of prospective studies, however, have not been able to establish that duration of TLI is an important determinant of overall laryngeal injury.^{19,35,52,128,138,283,284} Rashkin and Davis found no correlation between acute laryngotracheal injury and duration of TLI in sixty-one adult patients intubated for more than 3 days,³⁵ and Kastanos et al found no such relationship in nineteen patients intubated longer than 1 day.¹²⁸ In a prospective study of trauma patients, Dunham and LaMonica found no difference in laryngotracheal pathology between those patients undergoing early tracheotomy and those with 2 weeks of TLI.²⁸⁴ Colice et al noted that serious laryngeal injuries could develop after only 1 to 3 days of TLI.⁵² In fifty-four patients surviving after TLI, they found no relationship between the frequency of nonstenotic laryngeal complications and the duration of intubation.⁵² Subsequently, Colice reported that prolonged TLI (>10 days) did not influence the rate of resolution of laryngeal injury.¹³⁸

Some animal studies support the absence of a relationship between duration of TLI and laryngeal complications. Bishop et al²⁸⁹ and Weymuller²⁹⁰ reported that length of intubation was not related to the severity of laryngeal injury in dogs. In the dog model used by these investigators, TLI caused mucosal inflammation and erythema by 24 hours and loss of mucosal architecture within 1 week, but laryngeal damage remained relatively stable after 1 week.²⁸⁹

It can be concluded that studies in both animals and humans fail to provide convincing evidence of an association between duration of TLI and laryngeal complications. Most prospective clinical studies have not been able to prove that duration of TLI by itself is an important determinant of laryngeal injury.

Age

Clinical and autopsy findings in prospective studies suggest that age is not a significant determinant of overall laryngeal injury from TLI.^{19,52,139}

Gender

The role of gender in laryngeal injury from TLI is unclear. Hedden et al reported in a retrospective study that laryngeal injury at autopsy following TLI was more severe in women than in men.²⁸⁵ Postmortem observations in separate studies could not confirm this relationship.^{19,131} Clinical studies have suggested that laryngeal stenosis following TLI is more common in women than in men.^{242,270} Lindholm interpreted his observations to suggest that tube size relative to laryngeal size is an important determinant of laryngeal injury,¹³⁰ but Colice et al reported contrasting results.⁵²

Hypotension

The pivotal role of laryngeal and tracheal capillary perfusion pressure in the pathogenesis of ETT-induced airway injury has led investigators to speculate that hypotension is an important risk factor for laryngeal injury during TLI. There are no convincing data, however, to support this concept.⁵²

Route of Intubation

Two studies suggest that posterior laryngeal ulcers are more common with oral intubation than nasal intubation.^{19,134} Possible reasons for this observation are that nasotracheal tubes are generally smaller than orotracheal tubes, they move less because they are stabilized by the walls of the nasal cavity, and they enter the glottis at a straighter angle of entry, thereby decreasing pressure on the posterior glottis. Another study, however, did not find less laryngeal injury with nasal intubation than oral intubation.²⁴³

Miscellaneous Factors

Underlying disease state, infection, corticosteroid therapy, head position, state of consciousness, gastric acid reflux, and many other factors have all been considered as possible determinants of laryngeal injury during TLI, but none of these factors have proven to have a contributing role. Several studies have suggested that laryngeal injury following TLI is more common in patients with diabetes mellitus.^{143,270} Bacterial invasion of subglottic tissue injured by TLI may have a role in the pathogenesis of subglottic stenosis. In a rabbit model of subglottic ischemic injury from ETT cuff overinflation, Kil et al observed diffuse mucosal and submucosal inflammation, necrosis, and loss of surface epithelium.²⁹¹ Treatment with high-dose dexamethasone 1 hour before extubation and 6 hours after extubation had no effect on the histologic pattern of injury.

Posterior glottic stenosis may be more likely if TLI is followed by tracheotomy.^{127,242} This association might occur because tracheotomy results in nonuse of the larynx, which might promote the formation of adhesions across ulcerated laryngeal surfaces.²⁴² Whited found a high statistical association between posterior laryngeal stenosis after TLI and subsequent tracheotomy, but cautioned that this might reflect selection bias rather than a true cause-and-effect relationship.^{126,127} Colice et al also reported in their prospective study that the severity of laryngeal injury after TLI was associated with subsequent tracheotomy.⁵²

In summary, it is likely that many factors contribute to laryngeal injury from TLI. Available data suggest that the mechanical force applied to the posterior larynx by the ETT is the most important risk factor for laryngeal injury. It is intuitive that abrasion of the laryngeal mucosa by movement at the interface with the ETT is an important risk factor, but supportive data are limited. Duration of TLI by itself has an uncertain role in the pathogenesis of laryngeal injury.

Clinical Influences in Laryngeal Granuloma Formation

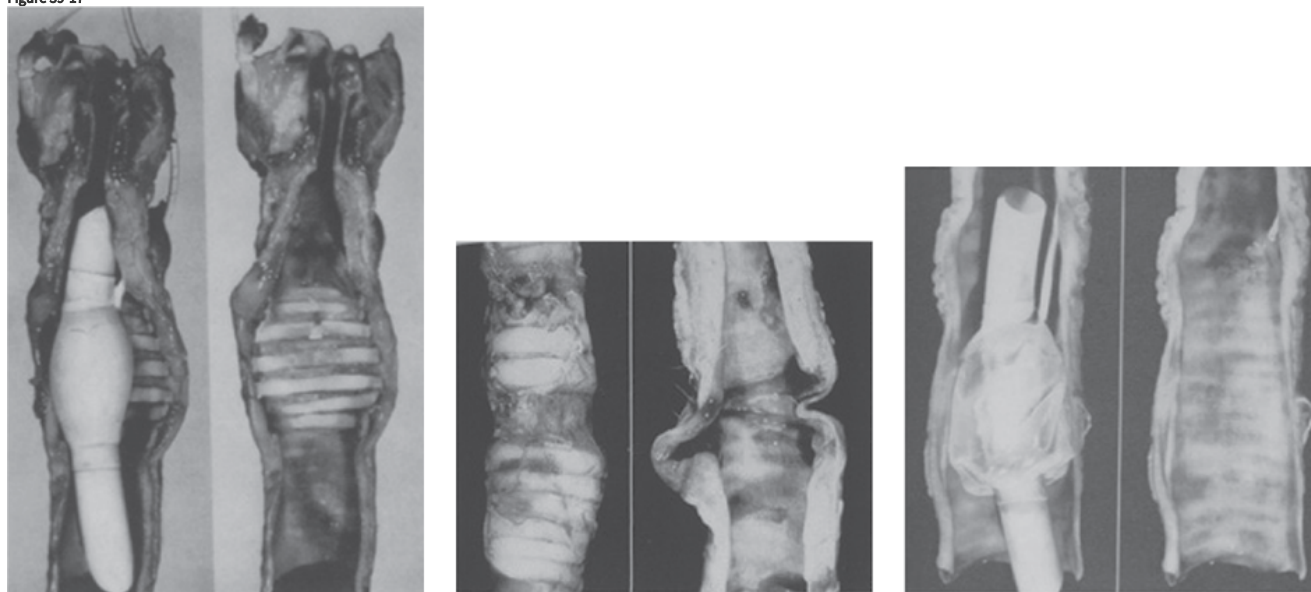
The prospective study of ninety-seven patients by Santos et al found significant associations between the early appearance of true vocal cord granulomas after extubation and (a) duration of TLI, (b) subsequent tracheotomy, (c) use of a larger ETT in those patients who required tracheotomy, and (d) use of a nasogastric tube.¹³⁹ The delayed appearance of true vocal cord granulomas 2 to 10 weeks after extubation was significantly associated with subsequent tracheotomy and with use of a nasogastric tube. Lindholm also found a positive correlation between ETT diameter and the subsequent development of laryngeal granulomas.¹³⁰

Clinical Influences in Tracheal Cuff-Site Injury

Cuff Pressure

The important role of high cuff inflation pressures in the pathogenesis of tracheal cuff-site injury is now well established, based on studies of both TLI and tracheotomy.^{148–150,254,256,258,260,278,279,292–296} The pioneering studies of Cooper and Grillo,^{148,278,279} Grillo et al,¹⁴⁹ Ching and Nealon,²⁹² Ching et al,²⁹³ and Nordin et al^{257,259,294} established conclusively in both animal models and in humans that hard cuffs were more injurious to the tracheal mucosa than soft cuffs. These observations were confirmed by other investigators.^{254,256,260} Figure 39-17 shows the critical role of cuff pressure in causing tracheal cuff-site injury in animal experiments performed by Cooper and Grillo.¹⁴⁸

Figure 39-17



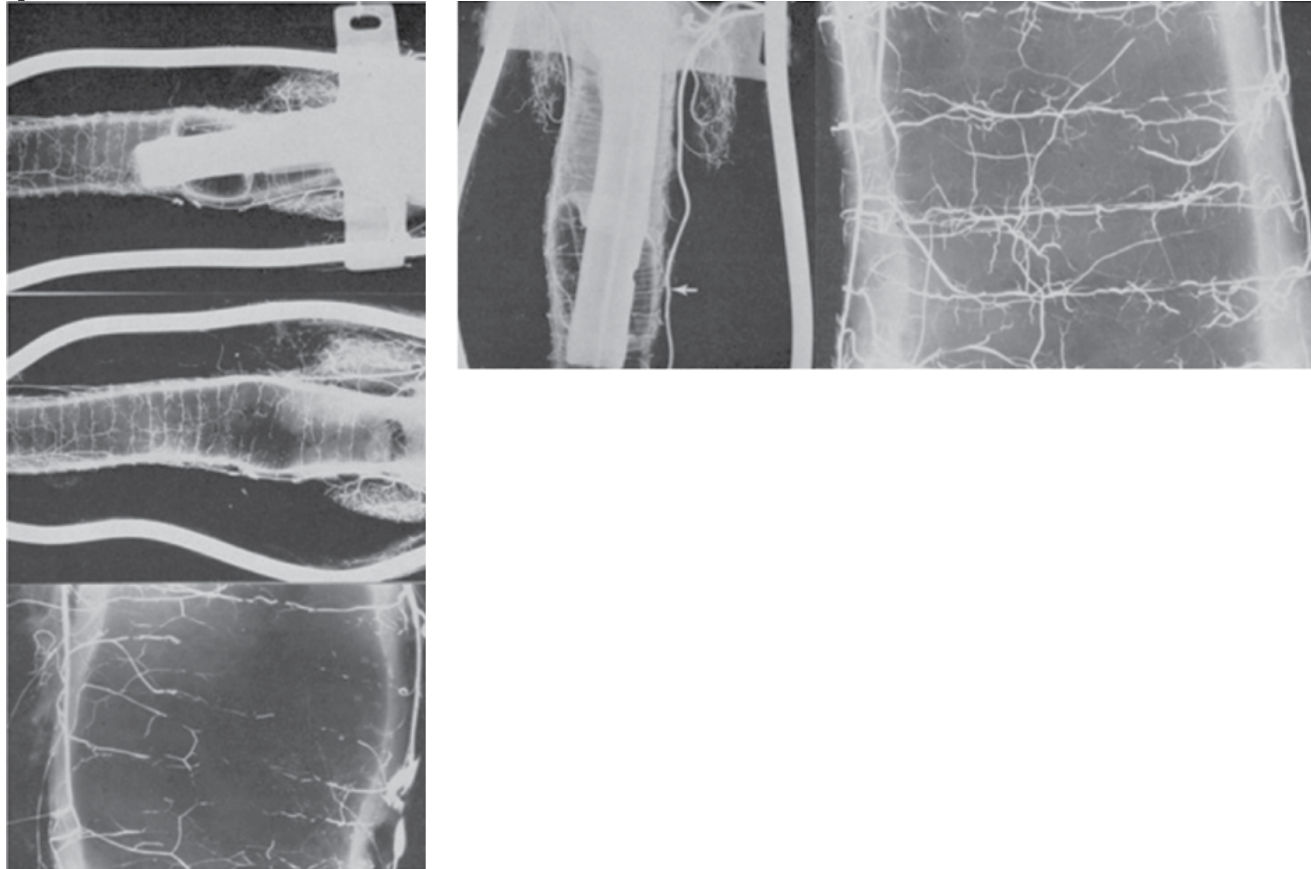
Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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An experiment in dogs indicated the critical role of high cuff-inflation pressure in causing severe tracheal cuff-site injury. Shown here are severe tracheal ulceration with exposed tracheal cartilage (*left*) and tracheomalacia (*center*) from prolonged intubation with endotracheal tubes (ETTs) with low-volume,

high-pressure cuffs. The panel on the *right* shows a normal tracheal surface after prolonged intubation with an ETT with a high-volume, low-pressure cuff. (Modified, with permission, from Cooper and Grillo.¹⁴⁸)

Intracuff pressure is transmitted laterally against the wall of the trachea (see Fig. 39-14). The intracuff pressure exceeds the pressure exerted against the tracheal wall (the lateral tracheal wall pressure) by a small amount in soft cuffs and by a greater amount in hard cuffs. Ischemia and eventual necrosis occur when the lateral tracheal wall pressure exceeds the capillary perfusion pressure of about 25 mm Hg,^{258–260,294,295} as illustrated in Figure 39-18. Ischemia of the tracheal mucosa occurs in rabbits when the cuff inflation pressure exceeds 30 mm Hg.²⁵⁷ Necrosis of the tracheal mucosa leads to sloughing and ulceration of the mucous membrane, exposing tracheal cartilage. Continued ischemia may be followed by partial or complete destruction of cartilaginous tracheal rings and loss of the structural integrity of the affected tracheal segment, leading to tracheal dilation (see Fig. 39-14).^{153,154} Healing of the injured tracheal segment during any stage of this process may lead to a tight fibrous stricture (tracheal stenosis).

Figure 39-18



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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Radiographs of the tracheal arteries in dogs were performed to investigate the effect of cuff pressure on blood flow from the tracheal arteries to the wall of the trachea at the cuff site. The panel with three images on the *left* shows a marked reduction in small arteries at the cuff site when a low-volume, high-pressure cuff of a tracheostomy tube was inflated for 18 hours to just-seal volume (*top*: cuffed tube in place; *center*: tube removed; *bottom*: magnified view of cuff site). The panel with two images on the *right* shows intact small arteries when a prestretched cuff having pressure–volume characteristics of a high-volume, low-pressure cuff was inflated in the same fashion (*left*: cuffed tube in place; *right*: magnified view of cuff site with tube removed). (Modified, with permission, from Dunn et al.²⁹⁵)

Modern soft-cuff ETTs transmit intracuff pressure to the tracheal wall, but not to the extent that hard-cuff ETTs do. Overinflation of soft cuffs, however, may cause high lateral tracheal wall pressures to develop, even to the point at which capillary perfusion pressure is exceeded.²⁵⁸ In the clinical setting, it is important to note that variations in ETT cuff pressure are commonly observed after the ETT pressure is set to a safe level, so high pressures may develop periodically.²⁹⁷ The amount of lateral tracheal wall pressure appears to be more important than the duration of intubation in producing tracheal damage.^{259,296}

Cuff Design and Shape

The geometric shape of the cuff relative to the shape of the tracheal lumen has been implicated in tracheal cuff-site damage.^{293,298} Asymmetric inflation of cuffs, leading to focal tracheal mucosal injury, was occasionally reported when hard-cuff ETTs were used, but with soft-cuff tubes that inflate evenly, this

phenomenon is no longer observed. Because modern soft-cuff tubes do have quite variable physical characteristics, including inflation diameter, fabric thickness, compliance, and geometric shape, all are not the same in their pressure–volume relationship and potential for mucosal injury.^{254,273,281}

Age

Prospective studies that have examined the influence of age on tracheal cuff-site injury have not found a conclusive relationship.^{19,128,129}

Gender

The role of gender in cuff-site tracheal injury from TLI is controversial, as prospective studies have found conflicting results.^{19,128,129}

Duration of Translaryngeal Intubation

The relationship between duration of TLI and tracheal cuff-site injury is controversial, probably because observational clinical studies have not been able to control for confounding variables such as cuff pressure, tube movement, and ETT size. We found no statistical association between tracheal cuff-site injury at autopsy and duration of TLI or duration of TLI and tracheotomy together.¹⁹ Other studies have also failed to find a significant relationship between duration of tracheotomy and cuff-site injury.^{129,299,300} Only one prospective study found that tracheal cuff-site injury was significantly associated with the duration of TLI.¹²⁸

Hypotension

Few studies have examined the relationship between hypotension and tracheal cuff-site injury in humans. We found no relationship between hypotension and tracheal cuff-site injury at autopsy.¹⁹ In an experimental study of dogs with cuffed tracheostomy tubes, Dunn et al found that intercurrent periods of hypotension did not affect the extent of tracheal injury from tracheostomy tube cuffs.²⁹⁵

Corticosteroid Therapy

Two prospective studies reported no association between corticosteroid therapy and tracheal cuff-site injury from TLI.^{19,128}

Airway Infection

Nordin reported that bacterial invasion of tracheal mucosa damaged by high cuff inflation pressures begins to occur about 4 hours after the onset of cuff inflation.²⁵⁹ There is, however, no convincing evidence from clinical or experimental studies that airway infection increases the risk of cuff-site complications.^{129,278}

Miscellaneous Factors

Kastanos found a statistically significant association between tracheal injury from TLI and the use of positive end-expiratory pressure.¹²⁸ Excessive motion between the inflated cuff and the trachea was implicated in tracheal injury by the studies of Lindholm.¹³⁰

Siobal et al reported a case of tracheoinnominate artery fistula in a burn patient who was intubated for 40 days with an 8-mm internal diameter ETT, which was equipped with a subglottic suction port and held in position by a wire suture to a molar tooth.¹⁶⁵ Erosion of the left anterior wall of the trachea by the tube tip was attributed to the prolonged period of intubation, fixed position of the tube, and decreased flexibility of the tube as a result of the suction channel feature.

Clinical Influences in Pulmonary Complications

Pneumonia and Tracheobronchitis

This topic is discussed in [Chapter 46](#). Myriad factors predispose to VAP and ventilator-associated tracheobronchitis while an ETT is in place,^{177,179,180,190,301} including the underlying disease(s); impaired host defenses; antibiotic exposure; aspiration; enteral feeding tubes; sinusitis; impaired mucociliary function; ineffective cough; respiratory equipment; hospital transport out of the ICU; ETT biofilm; and increased binding of bacteria to the tracheobronchial epithelium. During intubation, the lower airways of ICU patients are heavily colonized with a variety of organisms. Those isolated most frequently as pathogens in VAP are aerobic gram-negative bacilli (especially *P. aeruginosa*, *Enterobacteriaceae*, and *Acinetobacter* species) and *S. aureus*.¹⁹⁰ Polymicrobial infections in VAP are common. Anaerobic organisms may also be isolated in cultures of lower respiratory tract secretions in patients with VAP,³⁰² but their significance in the pathophysiology of VAP remains controversial.³⁰³

An indwelling ETT is by itself a risk factor for VAP and ventilator-associated tracheobronchitis, because it is a foreign body and it bypasses protective upper airway defenses. Aspiration of contaminated upper airway secretions and leakage of bacteria-laden fluid that pools above the inflated ETT cuff are important mechanisms of entry of pathogens into the lung.^{177,304,305} The ETT biofilm also serves as a reservoir for microorganisms, increasing the risk for VAP and ventilator-associated tracheobronchitis.^{274,306} Reintubation may be a separate important risk factor for VAP.³⁰⁷

Aspiration

Aspiration of gastric and upper airway secretions into the lower airway while an ETT is in place may occur for a variety of reasons.¹⁸⁹ TLI per se invites aspiration of secretions because of stenting of the larynx in an open position, prevention of glottic closure, esophageal compression by the inflated cuff, pooling of secretions above the cuff, creation of channels along folds of cuff fabric,²⁸⁰ and desensitization of protective upper airway reflexes. Elpern et al, in a prospective study of aspiration in intubated adults, observed that cuff inflation to occlusion does not prevent aspiration, nor does a head-up position in bed.¹⁸⁹ They also noted that nasogastric tubes increased the risk of aspiration, whereas level of consciousness and feeding status did not.

Complications during and after Extubation

In a prospective study of critically ill adults, Jaber et al found that postextubation stridor was associated with traumatic or difficult intubation, prior self-extubation, and increased cuff pressure at the time of ICU admission.²²⁵ They also observed that postextubation stridor increased with duration of TLI,²²⁵ a finding that was also observed by Tadie et al.²²¹ Efferen and Elsagr reported that postextubation stridor was associated with cuff pressure, corticosteroid therapy at the time of extubation, and a neurologic disorder necessitating TLI.²²⁶ Megarbane et al observed that female gender and intubation time were predictive of postextubation laryngeal injuries in acutely poisoned patients.²²⁰

The pathogenesis of laryngeal and tracheal stenosis following TLI is related to the healing process of airway injury incurred during TLI. Why some ulcerative lesions heal normally, others with granuloma formation, and others with cicatrix or scar formation is obscure. Table 39-9 summarizes important or potentially important mechanisms of laryngotracheal injury from TLI.

Table 39-9: Important Mechanisms of Laryngotracheal Injury from Translaryngeal Intubation

Laryngeal ulceration
High pressure on the posterior endolarynx
Curvature of the endotracheal tube (ETT)
Round shape of the ETT
Stiffness of the ETT
Abrasion (excessive movement at the tube-mucosa interface)
Large diameter of the ETT
Female gender
Oral route of intubation
Laryngeal granuloma formation
Laryngeal ulceration
Nasogastric tube
Large diameter of the ETT
Increasing age
Laryngeal stenosis
Laryngeal ulceration
Subsequent tracheotomy
Laryngeal inactivity
Bacterial invasion
Tracheal cuff-site ulceration
High lateral tracheal wall pressure

Important Prospective Studies of Complications of Translaryngeal Intubation

Because the literature on complications of artificial airways is largely based on retrospective and anecdotal observations, study of this topic is especially prone to potential bias and faulty conclusions. Few techniques of airway management satisfy the criteria of evidence-based medicine. Fortunately, however, a growing number of prospective studies of complications of TLI, tracheotomy, or both are now available. Large prospective studies of complications of TLI are difficult to perform but they do provide valuable insights.

A growing number of studies comparing complications of TLI with those of tracheotomy or complications of delayed versus early tracheotomy have employed a randomized design.^{129,284,308–313} Prospective studies attempting to randomize patients to early tracheotomy have been difficult to perform. Only a few prospective studies have attempted to directly compare complications of TLI alone with those of tracheotomy.^{19,129,130,284} A growing number of prospective studies have evaluated airway complications of early versus delayed tracheotomy,^{308,309,311–315} but the complications that can be attributed to TLI alone in these studies are difficult to assess.

Rumbak et al randomized 120 adult medical ICU patients to early percutaneous tracheotomy within 48 hours or delayed tracheotomy after 10 to 14 days of TLI.³⁰⁹ The early tracheotomy group experienced significantly lower mortality, less-frequent pneumonia, fewer days in the ICU, fewer days of mechanical ventilation, and fewer days of sedation. Sugerman et al, however, using a similar study design, found no significant differences between the early and the delayed tracheotomy groups with regard to mortality rate, days in the ICU, or frequency of pneumonia.³⁰⁸ This section reviews selected prospective clinical studies of laryngeal or tracheal complications of TLI in adults (Table 39-10).

Table 39-10: Selected Prospective Studies of Complications of Translaryngeal Intubation

Year	References	No. of Patients
1969	Lindholm ¹³⁰	457
1981	Stauffer et al ¹⁹	150
1984	Whited ¹²⁷	200
1989	Colice et al ⁵²	82
1994	Santos et al ¹³⁹	97
1995	Schwartz et al ⁷⁰	238
2004	Bouderka et al ³¹⁰	62
2006	Jaber et al ²⁰	220
2007	Benedetto et al ³¹⁷	150
2008	Griesdale et al ²¹	136

Lindholm's study represents the most comprehensive study of complications of TLI ever performed.¹³⁰ He investigated 457 patients for acute and late complications of intubation. Very few late, severe complications and no deaths were attributed to TLI. Large ETT size, ETT stiffness, and excessive laryngeal motion were identified as important risk factors for laryngeal injury. Because, however, the patients were intubated in large part with red rubber and latex tubes with hard cuffs, and the duration of intubation was less than 3 days, the results of the study are not entirely applicable to modern critical care.

Stauffer et al compared complications of TLI with those of tracheotomy in 150 critically ill adults in two large teaching hospitals.¹⁹ Survivors were studied for late complications and nonsurvivors were evaluated at autopsy. An attempt to randomize patients into early tracheotomy and prolonged TLI groups was only partially successful. Approximately two-thirds of TLI and two-thirds of tracheotomies had one or more complications, but the complications were judged to be more severe in the tracheotomy group. Laryngotracheal injury at autopsy did not correlate with the duration of TLI. A few patients appeared to tolerate TLI well for up to 3 weeks.

Whited described his personal experience as an otolaryngologist-head and neck surgeon with 200 patients from the critical care units of two teaching hospitals who had TLI from 2 to 24 days.¹²⁷ Mirror or fiber-optic laryngoscopy tracheoscopy was performed at extubation and serially thereafter. Fifty survivors (group I) had TLI for 2 to 5 days, 100 (group II) had TLI for 6 to 10 days, and fifty (group III) had TLI for 11 to 24 days. The severity of posterior laryngotracheal injury appeared to increase with the duration of TLI, but statistical analysis was not provided. Chronic posterior commissure stenosis progressing inferiorly to the level of the cricoid ring and into the trachea occurred in 14% of group III patients. Chronic stenosis was more common in patients with TLI followed by tracheotomy than in those with TLI alone. Other variables that might have affected the outcome were not described. The author concluded that TLI for less than 7 days is safe and that TLI beyond 10 days is unacceptable as a routine policy.

Colice et al examined eighty-two adult patients intubated more than 4 days for laryngeal complications of TLI using direct fiber-optic laryngoscopy and careful follow-up.⁵² Nonlaryngeal complications were not evaluated. Ulceration and edema of the posteromedial aspects of the vocal cords were observed in 94% of the patients at extubation. Neuromotor activity and performance of a tracheotomy, but not duration of TLI, were associated with increased laryngeal complications. Initial findings at laryngoscopy did not predict the late development of adverse effects.

Santos et al reported risk factors for laryngeal injury from prolonged TLI in critically ill adult men in an initial report in 1989,³¹⁶ and updated their findings with a second report in 1994.¹³⁹ The latter study described prospective observations in ninety-seven adult men with orotracheal TLI (mean duration: 9 days). Postextubation endoscopic findings in seventy-nine survivors included laryngeal erythema (94%), laryngeal ulceration (76% with resolution within 6 weeks), and laryngeal granulomas (44%). True vocal cord immobility was seen in sixteen (20%) patients and was delayed in onset in half the cases. Duration of TLI was significantly associated with true vocal cord granuloma formation and early and delayed true vocal cord immobility but not with true vocal cord ulceration. The use of a larger ETT (size 8.0-mm internal diameter as compared with size 7.5 mm) was significantly associated with true vocal cord erythema, ulceration, granuloma formation, and delayed immobility. None of sixty-two patients with TLI alone developed laryngotracheal stenosis.

Schwartz et al prospectively studied 297 consecutive TLIs in 238 critically ill adults in medical and surgical critical care units of a large teaching hospital so as to assess immediate complications of TLI.⁷⁰ The most common indications for TLI were respiratory failure (50%), airway protection (17%), ETT change (13%), and cardiac and/or respiratory arrest (10%). Eleven percent of TLIs required three or more attempts at ETT placement. Eight percent were classified as difficult, but none of these were associated with adverse outcomes. Esophageal intubation occurred during twenty-five (8%) of the 297 intubation procedures and was not accompanied by any adverse sequelae. A new infiltrate on the postintubation chest radiograph was observed in 8% of cases, and in half of these (4%) aspiration was suspected as the cause.

Bouderka et al reported results of a prospective, randomized trial of early tracheotomy (days 5 to 6) versus prolonged TLI in sixty-two adults with severe head injury.³¹⁰ The two groups were similar in terms of age, gender, and simplified acute physiology score. Clinical symptoms of laryngotracheal complications did not differ between the two groups, but routine endoscopy to evaluate patients for these complications was unfortunately not performed. The early tracheotomy group experienced fewer days of mechanical ventilation both overall and after pneumonia was diagnosed. No differences in ICU stay, frequency of pneumonia, or mortality were observed.

Jaber et al prospectively evaluated immediate complications of 253 intubations in 220 patients in seven ICUs of two teaching hospitals.²⁰ Seventy-one (28%) of the intubations generated one or more severe complications, including severe hypoxemia (26%), hemodynamic collapse (25%), and cardiac arrest (2%). The presence of acute respiratory failure and shock were independent risk factors for intubation complications, and resident supervision in intubation reduced the risk of complications.

Benedetto et al performed a prospective observational study of urgent TLI in 150 patients in general hospital units at the Massachusetts General Hospital.³¹⁷ The overall rate of complications of emergent TLI was 27%, compared with a rate of 22% in elective TLIs. The most common complications observed were multiple (three or more) attempts at TLI and esophageal intubation, both occurring in 9% of cases. These event rates were strikingly similar to those observed by Schwartz et al⁷⁰ in another large teaching institution (11% for multiple attempts at TLI and 8% for esophageal intubation).

Griesdale et al examined early TLI complications in 136 patients in a prospective study in a single Canadian teaching hospital ICU.²¹ Nearly all intubations were supervised, but 39% generated one or more complications. Like Jaber et al,²⁰ Griesdale and coworkers observed that severe hypoxemia and hypotension were the most common severe complications of TLI. All intubations were successful, but 13.2% required three or more attempts.

Recognition and Management of Complications of Translaryngeal Intubation

Recognition of Selected Complications

Complications during Endotracheal Tube Placement

Right Main-Stem Bronchus Intubation

The classic physical examination findings of a patient with right main-stem bronchus intubation (see Fig. 39-3) are greater intensity of breath sounds and expansion of the right hemithorax compared with the left. Right main-stem intubation, however, may be difficult to recognize in many patients, especially those with emphysema, because of distant breath sounds on auscultation and chest hyperinflation.³¹⁸ Schwartz et al observed that neither physical examination nor referencing the centimeter markings on the ETT shaft indicated malpositioning of the ETT after emergency TLI.⁶³ Women are at greater risk than men for malpositioning of the ETT in critical care intubations.⁶³ Right main-stem bronchus intubation may lead to early hypoxemia and cyanosis, particularly if the patient has underlying lung disease.

Esophageal Intubation

Inadvertent intubation of the esophagus should be recognized immediately by capnography (see the section [Prevention of Complications of Translaryngeal Intubation/Complications During Endotracheal Tube Placement/Avoiding Esophageal Intubation](#), and [Chapter 38](#)), although this method is not foolproof. Esophageal intubation may be recognized when manual ventilation after intubation produces poor chest expansion and gurgling sounds on auscultation over the stomach. In some cases, esophageal intubation may be very difficult to detect on physical examination. The patient with an ETT placed in the esophagus may display some degree of chest expansion, and “pseudo” breath sounds may be auscultated as gas enters the stomach. Esophageal intubation should be recognized before a postintubation chest radiograph is obtained. If a radiograph is obtained, however, it may reveal displacement of the ETT slightly away from the tracheal air column. Caution is necessary, however, because slight rotation of the patient may superimpose the tracheal air column over the ETT positioned in the esophagus.

Complications While the Endotracheal Tube Is in Place

Sinusitis and Otitis

A high index of suspicion is critical to establishing the diagnosis of TLI-associated sinusitis and initiating effective therapy. Sinusitis may present as sinus pain and tenderness, or purulent nasal sinus drainage in an intubated patient, but it is important to emphasize that these findings may be minimal or absent. Sinusitis complicating nasotracheal intubation or use of nasogastric tubes typically presents with fever, or in some patients, with bacteremia and sepsis syndrome.

Most reports have emphasized the value of CT scanning for accurate diagnosis of intubation-associated sinusitis, because physical examination is generally unreliable and standard sinus radiographs often fail to establish the diagnosis, especially when there is involvement of the ethmoid or sphenoid sinuses.³¹⁹ Radiographic maxillary sinusitis is defined as the presence on CT scanning of an air-fluid level or opacification of the maxillary sinus. Opacification of the other sinuses is common in intubated patients. Otitis is detected by routine otoscopic examination, which should also be performed in evaluation of fever of unknown origin in the intubated patient.

Nasal and Oral Injury

Nasal and oral complications during TLI are recognized by daily careful examination of the nose and mouth.

Laryngeal Injury

Recognition of laryngeal injury during TLI is rarely successful, because the ETT hides the injured areas from observation. Posterior ulceration of the larynx is hidden from view during fiber-optic endoscopy unless the ETT is removed, which is dangerous and impractical. Supraglottic and glottic edema is sometimes assessed by direct or indirect laryngoscopy during TLI. There are conflicting reports about the value of laryngoscopy in assessing laryngeal injury during TLI.^{140,143} Routine endoscopy of the pharynx and larynx during TLI is not currently advised. Furthermore, endoscopic findings in the larynx during TLI do not clearly guide the timing of tracheotomy or predict long-term laryngeal sequelae of TLI.^{52,246}

Tracheal Injury

Tracheal cuff-site injury cannot be detected during TLI unless fiber-optic bronchoscopy is performed while the ETT is retracted upward. This practice has no proven value, however, and is not indicated in decision making about extubation or conversion to tracheotomy. When the ratio of cuff-site diameter to tracheal transverse diameter on a chest radiograph exceeds 1.5, severe damage to the tracheal wall is present.¹⁵² Eighteen (13.5%) of 135 intubated adult patients developed this complication as a result of cartilaginous destruction, and those who survived eventually developed tracheal stenosis or tracheoesophageal fistula.¹⁵² Therefore, patients in whom tracheal dilation develops during TLI should be followed closely for months because of the potential for tracheal stenosis.

Complications during and after Extubation

Laryngeal Injury

Persistent hoarseness, stridor, or dyspnea after extubation should always raise the question of serious laryngeal complications of TLI, including granuloma formation, laryngeal muscle paresis or paralysis, severe ulceration, laryngeal stenosis, cricoarytenoid ankylosis, or synechia formation.^{125,130,242} The physician should also be aware that dyspnea, hoarseness, and stridor appearing weeks to months after extubation may indicate serious laryngeal (or tracheal) stenosis and should not be dismissed as simply being related to the underlying lung disease.²⁴⁰ Glottic or subglottic stenosis may take weeks or months to develop after seemingly uneventful TLI and extubation.

The evaluation of patients with suspected serious laryngeal injury after TLI should include physical examination and laryngoscopy. Physical examination may reveal prolonged inspiratory phase, harsh tracheal sounds and palpable thrill over the larynx on inspiration, and inspiratory intercostal retractions. Upper airway stridor transmitted to the thorax should not be misinterpreted as a sign of bronchoconstriction.

Laryngoscopy is necessary in all cases. Vocal cord movement may be evaluated initially by mirror laryngoscopy. If any abnormalities are detected, direct laryngoscopy should be performed. The finding of bilateral vocal cord abduction could be related to cricoarytenoid arthritis or recurrent laryngeal nerve injury.²⁴² Recurrent laryngeal nerve injury is suggested by absence of vocal cord motion, while in posterior glottic stenosis there is some motion of the cords on inspiration and phonation.²⁴² Glottic synechiae and laryngeal granulomas are usually readily apparent at the time of direct laryngoscopy.

Imaging the larynx is very helpful in the diagnosis of TLI-induced laryngeal injury. Older imaging techniques such as soft-tissue radiographs and linear tomography have been replaced by CT scanning with multiplanar reconstruction capability to create axial, coronal, and sagittal reformatted images.³²⁰ Magnetic resonance imaging plays a lesser role than CT imaging. Radiographic studies and fiber-optic laryngoscopy are more sensitive than the history and physical examination in detecting laryngeal injury from TLI.

The maximal expiratory-to-inspiratory flow-volume loop is sometimes performed in the evaluation of patients with suspected laryngeal (or tracheal) injury from TLI. The classic feature of unilateral vocal cord paralysis on a flow-volume loop is attenuation of inspiratory flow rates. The classic feature of bilateral vocal cord paralysis or laryngeal (or tracheal) fixed stenosis is attenuation of both inspiratory and expiratory flow rates with a “boxed-off” appearance to the flow-volume loop (see Fig. 39-13A). Classic patterns of these types of injury are not always seen, however, because the flow-volume loop is neither sensitive nor specific. A completely normal flow-volume loop, on the other hand, makes functionally significant laryngeal (or tracheal) disease unlikely.

Tracheal Stenosis

Symptomatic tracheal stenosis after extubation is very uncommon in the modern era of soft-cuff ETTs. Like laryngeal stenosis, tracheal stenosis may not become apparent for weeks, months, or even years after extubation.³²¹ Typical symptoms of tracheal stenosis include progressive dyspnea on exertion, cough, difficulty clearing sputum, and in some cases, stridor. Unfortunately, these findings are commonly attributed to underlying disorders such as bronchitis, asthma, chronic obstructive pulmonary disease, or resolving acute respiratory distress syndrome.²⁴⁰

The diagnosis of tracheal stenosis requires a high degree of clinical suspicion based on a history of TLI (or tracheotomy) weeks to months earlier. Initial symptoms of cough, hoarseness, and difficulty clearing lower airway secretions are followed by dyspnea and stridor. Stridor and progressive respiratory distress, however, may not occur until the tracheal diameter is reduced to approximately 5 mm or less in adults.²⁴⁷ Physical examination may be normal in patients with mild to moderate degrees of tracheal stenosis. In severe cases, stridor is the key finding. Flow-volume loops demonstrate fixed airway obstruction if there is a tight fibrous ring at the stenotic segment (see Fig. 39-13A). A 50% reduction in tracheal caliber (<8-mm diameter), however, may be needed in order to produce an abnormal flow-volume relationship.³²²

Confirmation of the diagnosis of tracheal stenosis is possible with multidetector CT scanning, including axial imaging, multiplanar reconstruction, and creation of “virtual bronchoscopy” images, all of which are highly accurate in comparison with fiber-optic bronchoscopy.^{251,323} Multislice three-dimensional CT imaging offers remarkably clear images of tracheal stenosis.³²⁴ Fiber-optic or rigid bronchoscopy may be employed in some cases.

Management of Selected Complications

Complications While the Endotracheal Tube Is in Place

Sinusitis

Effective management of sinusitis during TLI requires early recognition, appropriate radiographic studies (see “Complications During and After Extubation” above), and culture and sensitivity of aspirated sinus fluid. Nasotracheal and nasogastric tubes should be removed. Consultation with an otolaryngologist-head and neck surgeon is advised. Aspiration of maxillary sinus fluid yields fluid for Gram stain and cultures for bacteria (aerobic and anaerobic) and fungi, allowing selection of appropriate antimicrobial therapy. Repeated aspiration may be necessary in some patients. Sinus irrigation or lavage via an indwelling catheter is sometimes required. Treatment with nasal decongestants for short periods is commonly employed.

Complications during and after Extubation

Laryngeal Complications

Laryngeal edema from mucosal trauma by the ETT may cause stridor, which appears minutes to several hours following extubation. Kriner et al prospectively observed that twenty (4.3%) of 462 patients who were intubated for more than 24 hours developed postextubation stridor.³²⁵ Female gender, longer duration of TLI, and larger ratio of ETT diameter to laryngeal size were risk factors for postextubation stridor in this series. Fundamental elements of management of postextubation stridor include supplemental oxygen, close monitoring of vital signs and oxygen saturation, removal of secretions from the upper airway, and positioning the head and neck to maintain airway patency. The patient should be reassured and anxiety managed appropriately. Because most cases of postextubation stridor are caused by laryngeal edema, management should be directed with this in mind. Parenteral corticosteroids, aerosolized epinephrine, and inhalation of a helium-oxygen gas mixture are the mainstays of empiric therapy.¹⁴⁴ Helium-oxygen gas mixtures may help reduce the patient's work of breathing. Helium-oxygen mixtures with more than 40% oxygen are generally ineffective. Reintubation is commonly required if the above measures are not successful. Kriner et al noted that seven (35%) of twenty patients with postextubation stridor required reintubation.³²⁵

Helium–oxygen therapy has also been described for temporary relief of increased work of breathing in patients with bilateral vocal cord dysfunction complicating TLI.²³⁰ Prevention of recurrent postextubation stridor is discussed (see the section [Prevention of Complications of Translaryngeal Intubation/Complications During and After Extubation/Avoiding Stridor](#)).

The cuff-leak test is valuable in planning for extubation, because little or no leak around the deflated cuff helps to predict the presence of laryngeal edema and postextubation stridor.³²⁶ Jaber et al observed that a cuff-leak volume of less than 130 mL (or less than 12% of tidal volume set at 10 to 12 mL/kg) was useful in identifying critically ill adult patients at risk for stridor after extubation.²²⁵ Others have reported that cuff leaks of less than 10%²⁸⁸ or 15.5%³²⁷ of tidal volume help to identify patients at risk for reintubation from laryngeal edema. Chung et al reported the sensitivity and specificity of the cuff-leak test to be 88.6% and 90.0%, respectively.²²³ Recent reports indicate that the negative predictive value of the cuff-leak test (air leak is observed) is 96% to 97%, while the positive predictive value (air leak is not observed) ranges from 12% to 25%.^{325,328,329}

As stated earlier, limited motion of the vocal cords long after extubation reflects posterior glottic fibrosis or ankylosis of the cricoarytenoid joint.¹⁰⁶ The goal of management is restoration of the voice, prevention of aspiration, and provision of adequate airflow. Management by an experienced otolaryngologist–head and neck surgeon is essential. Surgical and nonsurgical techniques to improve laryngeal function in patients with vocal cord paresis/paralysis are available.^{106,330–332} Patients with bilateral vocal cord paralysis may require emergent tracheotomy.¹⁰⁶

Treatments for laryngeal granulomas (see [Figs. 39-9B and C](#) and [39-12](#)) are directed at their causes, which include vocal abuse and gastroesophageal reflux disease as well as TLI. Granulomas from TLI are often managed with fiber-optic laryngeal surgery³³³ or surgical resection with very good results. Other therapeutic approaches for laryngeal granulomas have been described, including inhaled corticosteroids,³³⁴ injection of corticosteroids,¹⁰⁶ injection of botulinum toxin,³³⁵ and radiation.³³⁶ Wang et al, however, observed that vocal process granulomas, regardless of etiology, may completely resolve without steroid treatment and spontaneously achieve complete remission in more than 80% of cases without excision or other intervention.³³⁷

Laryngeal stenosis requires surgical intervention. Surgical options depend upon the location of the stenotic lesion and include laser therapy,³³⁸ dilation procedures, stenting, laryngofissure, resection,^{339,340} keel insertion, and other reconstructive procedures.^{341,342} Posterior commissure scarring is difficult to treat surgically.^{238,242} Repeated surgical operations for laryngeal stenosis may be required. In many cases, permanent tracheostomy is necessary. Czigner et al treated twenty-nine patients with subglottic stenosis, which was related to TLI or tracheotomy in most cases, with resection of the stenotic segment and end-to-end anastomosis, achieving a success rate of 96%.³⁴³ Pena et al reported very similar results.³⁴¹

Tracheal Complications

Tracheal granulomas are managed by endoscopic removal. Small granulomas may resolve spontaneously. Management of tracheal stenosis depends upon its severity. Patients with mild degrees of tracheal stenosis (arbitrarily defined as <25% reduction in airway diameter) and no symptoms require only periodic follow-up to look for progressive stenosis. Those with greater degrees of airway narrowing and symptoms require more aggressive management. In severe cases with 50% to 75% narrowing of the trachea, emergent medical and surgical intervention is necessary.

Surgical options in patients with severe tracheal stenosis include dilation procedures,^{344,345} stent placement, electrocautery, laser photoablation,³³⁸ and resection of the stenotic segment with reanastomosis.^{248,346,347} Tracheal reconstruction for tracheal stenosis is usually successful. Grillo and Donahue reported good or satisfactory results with surgery in 94% of 503 patients with postintubation tracheal stenosis.³⁴⁷

Prevention of Complications of Translaryngeal Intubation

Most of the complications of TLI are preventable. Skilled intubation technique, seasoned clinical judgment, and meticulous care and monitoring of the intubated patient are the key elements in avoidance of TLI complications. Because intubation and reintubation put a patient at risk for airway injury, noninvasive alternatives to TLI, such as use of the laryngeal-mask airway or the esophageal-tracheal Combitube, should always be considered, particularly in difficult and emergency airway management situations.^{348,349} A review of three randomized or controlled clinical trials of emergency TLI in the field demonstrated no benefit of TLI compared with bag-valve-mask ventilation techniques on survival or neurologic outcome.³⁵⁰ In patients with respiratory failure from acute exacerbation of chronic obstructive pulmonary disease, noninvasive positive-pressure ventilation decreases the need for TLI and has advantages, such as decreased mortality, rate of complications, and length of hospital stay.³⁵¹ An early important step is choosing the appropriate route of TLI for each patient with awareness of the advantages and disadvantages of each.³⁵² Principles of airway management to prevent complications of artificial airways are described in [Chapter 38](#) and elsewhere.^{120,185,353,354}

Complications during Endotracheal Tube Placement

Most of the complications that occur during ETT placement can be prevented by assiduously following the principles and procedures of airway management described in [Chapter 38](#). The intubator should be aware of common pitfalls in attempting to place an ETT in proper position ([Table 39-11](#)).

Table 39-11: Common Pitfalls in Performing Translaryngeal Intubation

Inadequate training, practice, and experience in intubation
 Failure to assemble necessary intubation equipment and drugs before starting the procedure
 Selection of the incorrect route of intubation
 Inadequate sedation, muscle relaxation, or topical anesthesia
 Failure to prepare the nose with vasoconstrictors, analgesics, and/or decongestants before nasal intubation
 Failure to oxygenate and ventilate the patient with bag-and-mask ventilation before intubation
 Improper positioning of the head and neck
 Use of the laryngoscope blade as a lever to pry the airway open
 Use of the teeth as a fulcrum to turn the laryngoscope
 Failure to visualize the vocal cords
 Forcing the endotracheal tube (ETT) forward against tissue resistance
 Prolonged intubation attempts without regard for progressive hypoxemia and acidosis
 Advancing the ETT deep into the airway after successfully passing the tube through the larynx
 Equipment malfunction

Source: This article was published in *Cin Chest Med*, Vol.12, Stauffer JL, Medical management of the airway, pages 449–482, Copyright © Elsevier 1991.

Selecting the Correct Endotracheal Tube

The ETT selected should have a high-volume, low-pressure soft cuff. All ETTs used in modern critical care should meet standards required by the F29 Committee on Anesthetic and Respiratory Equipment, American Society for Testing and Materials.²⁶⁷ The ideal ETT should be compliant at body temperature to help conform to the anatomic shape of the patient's airway, thereby reducing force on the posterior larynx. It also should be kink resistant and have a soft cuff that seals the airway and prevents aspiration at minimal lateral tracheal wall pressures.^{263,281}

For anesthetic intubation, Stenqvist et al have advocated the use of size 6-mm or 7-mm (i.e., the size of the internal diameter) ETTs in order to reduce pressure on the laryngeal contact points.³⁵⁵ For use in the patient with respiratory failure in the ICU, however, small tubes have obvious disadvantages, including difficulty in suctioning and increased airflow resistance that may increase the work of breathing and delay weaning from mechanical ventilation.³⁵⁶ A size 8.0-mm to 8.5-mm tube in men or a size 7.5-mm to 8.0-mm tube in women is usually selected for oral intubation, while a tube that is 0.5-mm to 1.0-mm smaller is usually selected for nasal intubations.¹⁸⁵

Preventing Intubation Trauma

Intubation trauma may be avoided by skilled airway management strategies (see [Chapter 38](#)). These include the initial use of bag-and-mask technique to oxygenate and ventilate the patient, thereby avoiding the necessity for “crash” intubation. The importance of adequate preoxygenation before attempting TLI cannot be overemphasized.^{357,358} Attempts to visualize the glottis with the laryngoscope should be brief and limited in number. Mort reported that the relative risk of airway complications was significantly higher when three or more laryngoscopic attempts were made.⁷² Eleven percent of the 297 TLIs in critically ill adults prospectively studied by Schwartz et al required this number of attempts.⁷⁰

The value of skill and experience in intubation cannot be overstated. For example, nearly all reports of pharyngeal perforation from TLI emphasize that this complication occurs in emergency intubations performed by inexperienced intubators. Skill in intubation may be gained by mannequin or simulation training.^{359–362} Bishop et al compared intubation by experienced intubators and novices, observing that novices required twice as long to intubate and had significantly higher intubation “impulses” (force of intubation multiplied by duration of effort) than experienced intubators.³⁵⁹ Schmidt et al documented a lower rate of complications when anesthetic TLI was supervised by attending physicians.²³ These observations underscore the benefits of training, skill, and experience in intubation technique.

[Chapter 38](#) discusses appropriate management strategies for the difficult intubation. Critical care providers should be familiar with current clinical practice guidelines to manage the patient with a difficult airway.³⁶³

Preventing Nasal Bleeding

Nasal intubation should be avoided in patients with coagulation disorders, thrombocytopenia, or platelet dysfunction. Careful examination of the nasal passage before intubation is essential, and the more patent side should be selected. Softening the nasal ETT with warm water before intubation may make it

more compliant, decreasing the frequency and severity of nasal hemorrhage.³⁶⁴ Application of a topical decongestant-vasoconstrictor and a topical anesthetic before nasotracheal intubation is recommended.

Preventing Hypoventilation and Hypoxemia

Monitoring of ventilation and oxygenation after intubation is critical (see [Chapter 38](#)). Unrecognized hypoventilation is among the most common preventable complications of intraoperative anesthesia. Recent guidelines emphasize the importance of preventing hypoventilation by confirming proper ETT placement (see the section [Avoiding Esophageal Intubation](#) and [Avoiding Right Main-Stem Bronchus Intubation](#))³⁴⁹ and by appropriately monitoring ventilation and oxygenation in the sedated patient.³⁶⁵ Guidelines of the American Society of Anesthesiologists indicate that all patients undergoing sedation/analgesia should be monitored with pulse oximetry and with observation or auscultation of ventilatory function.³⁶⁵ Monitoring of exhaled carbon dioxide should be considered for all deeply sedated patients and for moderately sedated patients whose ventilation cannot be directly observed.³⁶⁵

Preventing Cardiovascular Reactions to Translaryngeal Intubation

The tachycardic-hypertensive response and cardiac arrhythmias that may accompany laryngoscopy and TLI are potentially preventable.^{92,366} The routine use of drugs to prevent these cardiovascular reactions to TLI is not advised, but prophylactic treatment should be considered in patients with hypertension, cardiac disease, or cerebrovascular disease.³⁶⁶ A wide variety of drug classes and specific drugs are available for this purpose.³⁶⁶

Avoiding Esophageal Intubation

Esophageal intubation represents a misadventure in intubation with potentially disastrous consequences. A number of strategies have been described to avoid and to recognize this complication.^{10,367–370} McCulloch and Bishop reviewed the efficacy of fifteen different methods to determine correct ETT placement and avoid esophageal intubation.¹⁰ Pulse oximetry to assure adequate oxygenation and quantitative monitoring of carbon dioxide levels of expired gas are now standards of care in the operating room. No method to detect esophageal intubation is absolutely foolproof, but capnography comes closest to the mark.³⁶⁷ It is superior to chest auscultation^{370,371} and methods that employ a self-inflating bulb³⁷² or a lighted stylet.³⁷⁰ Monitoring of end-tidal CO₂ levels after intubation gives a characteristic repeated waveform if the ETT is properly located in the trachea. Failure of capnography to detect esophageal intubation is very rare.³⁷³ Although colorimetric CO₂ indicators are helpful in detecting esophageal placement of an ETT,^{368,369} they still have the potential for indicating tracheal placement even though the ETT is in the esophagus.³⁷⁴

Recent international guidelines for emergency cardiovascular care call for both primary confirmation of correct tracheal placement of the ETT by physical examination and secondary confirmation by an end-tidal CO₂ detector or an esophageal detector device.³⁴⁹ These steps are followed by careful chest auscultation and a chest radiograph. Use of an ETT holder to prevent subsequent dislodgment is also advised in these guidelines.³⁴⁹

Avoiding Right Main-Stem Bronchus Intubation

Physical examination of the chest is not reliable in determining proper positioning of the ETT.³⁷⁵ Brunel et al reported that 60% of patients with main-stem bronchus intubations had equal breath sounds on chest physical examination.³¹⁸ Therefore, in general, it is always desirable to obtain a chest radiograph after intubation^{318,376} unless fiber-optic bronchoscopy is performed to confirm proper positioning. By chest radiograph (or bronchoscopy) the tip of the ETT should be approximately 4 to 5 cm above the carina in adults, or at about the level of the third or fourth thoracic vertebra on the chest radiograph.^{377–379} When the tip of the ETT is approximately 4 cm above the carina, the upper end of the ETT cuff is approximately 2 cm below the vocal cords.³⁷⁹

Endobronchial intubation is usually caused by intubator inexperience and is therefore preventable with proper training and supervision.^{65,68,379} Intubators must resist the impulse to thrust the ETT too deeply into the trachea after the tube enters the glottis. Insertion to a depth of 21 cm in women or 23 cm in men from the upper incisor teeth or gums is a reasonable first step to avoid endobronchial intubation.^{65,68} This approach will probably avoid endobronchial intubation, but in some patients it may result in positioning the ETT too high in the trachea, risking laryngeal compression or accidental extubation. Cherng et al reported that the correct oral ETT length in adults with the head in a neutral position was correlated with body height.³⁷⁸ They advocated estimating the length of the tube (in centimeters) from 5 cm above the carina to the angle at the right side of the mouth with this equation: (body height in centimeters/5) – 13.³⁷⁸ After a chest radiograph confirms proper positioning of an ETT, the tube should be marked with indelible ink at the level of the lip or nasal alae to avoid subsequent inadvertent tube dislocation.

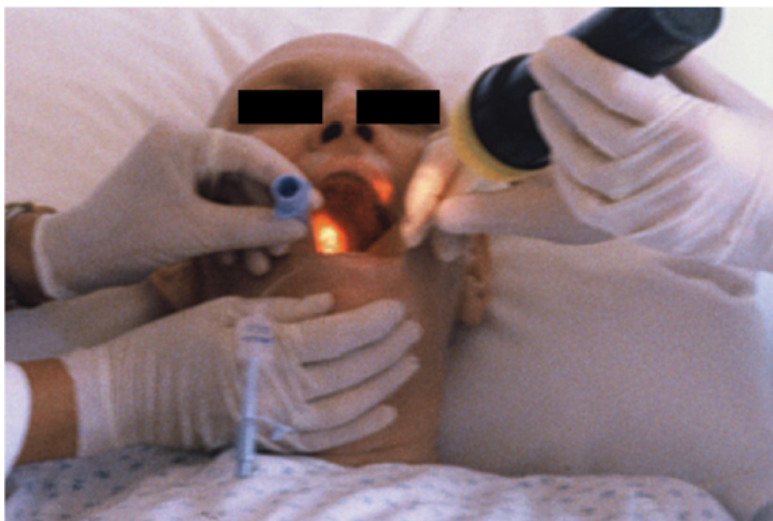
Complications While the Endotracheal Tube Is in Place

A number of practices may be helpful in preventing complications during TLI ([Table 39-12](#)). A careful physical examination of the patient should be performed by the physician at least daily, including examination of the mouth, nose, ears, sinuses, neck, and chest. The entry site of the ETT should be inspected carefully after removing adhesive tape or ETT holders as necessary ([Fig. 39-19](#)).

Table 39-12: Selected Measures to Reduce the Risk of Complications While the Endotracheal Tube Is in Place

Perform a careful bedside examination daily
 Include mouth, nose, ears, and sinuses
 Avoid nasal ulceration
 Position the nasotracheal tube so that it points downward
 Avoid too large a nasotracheal tube
 Avoid taping the endotracheal tube (ETT) tightly to the nose
 Avoid oral ulceration
 Minimize direct pressure on the lips and soft tissues of the oral cavity from the shaft of the ETT
 Support the weight of the attached ventilator tubing
 Use oropharyngeal airways only if necessary
 Select the best ETT for the patient
 Use high-volume low-pressure soft cuffs only
 Avoid very large ETTs (size ≥ 9.0 mm) to reduce the risk of laryngeal injury
 Manage cuff inflation carefully
 Limit intracuff pressure to ≤ 20 mm Hg, 25 mm Hg at the maximum
 Consider a trial of a foam cuff ETT if higher inflation pressures are required
 Check and record intracuff pressures every 8 hours
 Check for high cuff pressures after general anesthesia with nitrous oxide and other anesthetic gases
 Use the minimal occluding pressure technique for cuff inflation
 Avoid overinflation of soft cuffs
 Watch the chest radiograph for tracheal cuff-site dilation
 Avoid deliberate or inadvertent self-extubation
 Use extremity restraints and sedate the patient if necessary
 Reassure the patient to allay anxiety
 Be certain the ETT is properly located in the trachea, with the tip approximately 4 to 5 cm above the carina (for an adult of average height)
 Suction the airway carefully
 Avoid excessive movement of the ETT
 Use sterile gloved technique
 Oxygenate and ventilate the patient adequately during suctioning
 Limit suctioning to a maximum of 10 to 15 seconds at a time
 Suction secretions when necessary, not routinely
 Avoid transmission of mechanical forces directly to the patient
 Use elastic connectors and swivels
 Support the weight of ventilator tubing
 Do not tie ventilator tubing to the bedrails
 Minimize abrasion of the laryngeal and tracheal mucosa
 Minimize attempts to talk, swallow, and move the head and neck
 Control coughing and aggressively treat agitation, seizures, and rigid posturing
 Sedate the patient if necessary
 Use nasogastric tubes judiciously and only if necessary
 Avoid large nasogastric tubes
 Attach the ETT to a source of warm, humidified gas
 Give the patient reassurance and emotional support
 Facilitate and encourage written communication
 Remove the ETT as soon as possible

Figure 39-19



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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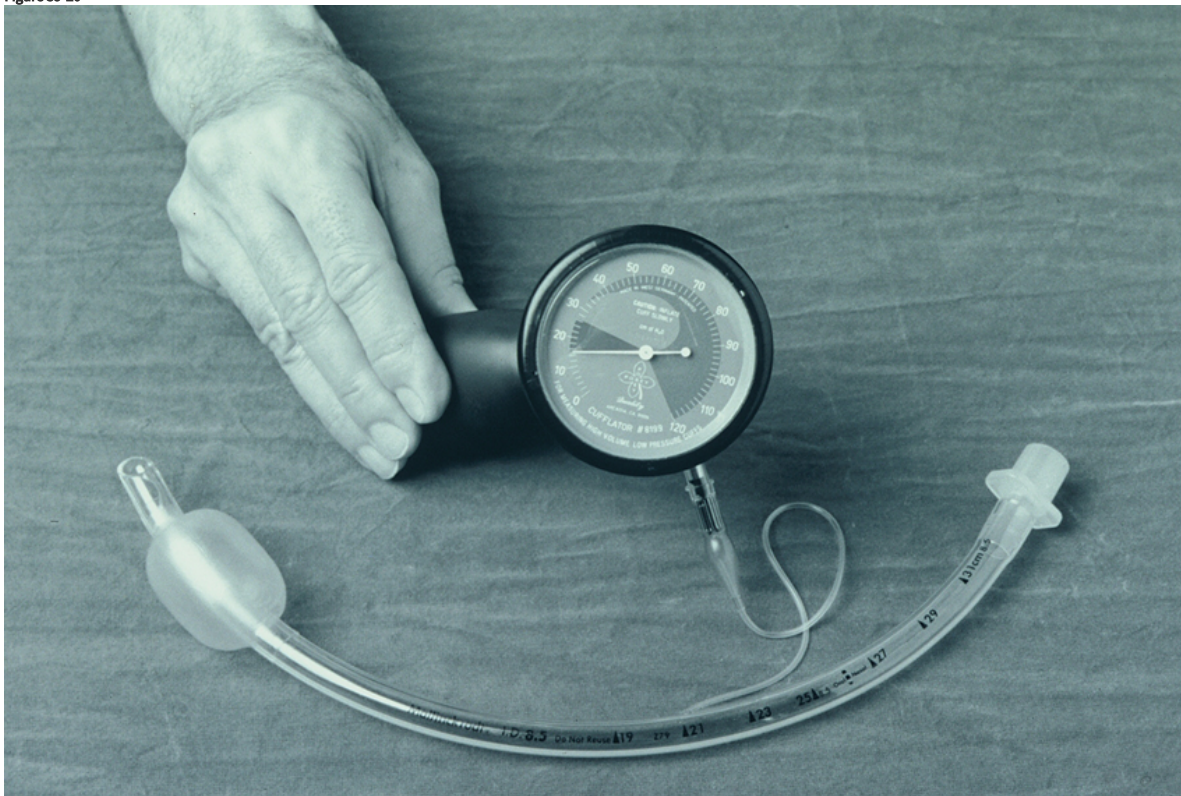
During transoral intubation the nose and mouth should be inspected carefully every day to detect evidence of nasal or oral injury from the endotracheal tube.

Nasal ala ulceration (see Fig. 39-4) may be prevented by positioning the ETT so that it points downward straight out the nose, by avoiding tight taping of the tube to the nose, by selecting a nasotracheal tube that is not too large for the patient, and by other approaches.^{380,381} Oral ulceration (see Fig. 39-7) may be avoided by minimizing the direct pressure transmitted to the lips and soft tissues of the oral cavity from the shaft of the ETT and ventilator tubing. Oropharyngeal airways should not be used routinely.

Managing Cuff Pressure

Only ETTs with low pressure cuffs should be used in adult critical care. The cuff inflation pressure should be limited to less than 20 to 22 mm Hg (27 to 30 cm H₂O) if possible.³⁸² If higher cuff pressures are required, the cuff pressure should certainly not be allowed to exceed 25 mm Hg (34 cm H₂O). If this goal cannot be achieved, consideration should be given to replacing the ETT with a foam-cuff tube. A cuff pressure monitoring device that records the pressure in the cuff and manometer simultaneously should be used (Fig. 39-20).

Figure 39-20



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The pressure in the endotracheal tube cuff should be set and checked every nursing shift to help avoid tracheal cuff-site injury. Cuff inflation pressure should be limited to less than 20 mm Hg (27 cm H₂O) if possible. The cuff pressure monitoring system shown here (Posey Cufflator Endotracheal Tube Inflator and Manometer, Posey Company, Arcadia, CA) records the pressure in the cuff and aneroid manometer simultaneously.

Critical care practitioners should be aware that it is necessary to check the ETT cuff for excessive pressure after general anesthesia with nitrous oxide (N₂O) and other anesthetic gases, because these gases diffuse into the cuff.³⁸³ In patients undergoing surgery with general anesthesia that included 50% N₂O, Tu et al noted that ETT soft-cuff pressures exceeded 80 cm H₂O and that cuff-site injury correlated with cuff pressure.³⁸³ ETT cuff pressure should be checked at least every 20 to 30 minutes during and after anesthesia with N₂O.

The “minimal occluding pressure” technique²⁷³ is advised for maintaining cuff inflation, although some practitioners prefer the “minimal leak” technique.³⁸⁴ Regardless of the technique of cuff inflation that is used, overinflation of soft cuffs should be avoided. The addition of only a few extra milliliters of air to the soft cuff may raise intracuff pressures sharply, increasing lateral tracheal wall pressure and risking tracheal mucosal injury.^{258,292,293} Cuff pressures should be recorded at least every 8 hours or once every nursing shift. Routine monitoring of intracuff pressures is desirable, despite the shortcoming that it always overestimates lateral tracheal wall pressure.³⁸⁵

High cuff inflation pressures should be taken seriously. The patient should be watched carefully for signs of excessive cuff pressure, including widening of the tracheal air column on the standard chest radiograph.¹⁵² Rising peak airway pressures during mechanical ventilation require higher cuff inflation pressures in order to prevent cuff air leak. A prospective study of fifteen patients undergoing mechanical ventilation for surgery revealed a linear relationship between peak airway pressure and minimal occluding cuff pressure, and in this series a cuff pressure of 25 mm Hg corresponded with a peak inflation pressure of 35.3 mm Hg (48 cm H₂O).³⁸⁶

Avoiding Unplanned Extubation

The patient should be constantly reassured to reduce anxiety. Extremity restraints should be applied if necessary and the patient sedated if it appears that self-extubation is likely. Mechanically ventilated patients who are severely agitated are significantly more likely to self-extubate than those who are not agitated.^{204,387} Tung et al observed in a retrospective case-control study that ICU patients who self-extubated were more than twice as likely as controls to be agitated.²⁰¹

Self-extubation tends to be repetitive, so constant vigilance is necessary. Staff awareness of the potential for self-extubation and the appropriate use of restraints and sedation reduce the risk of its occurrence. Proper positioning of the tip of the ETT approximately 4 to 5 cm above the carina also reduces the risk of inadvertent self-extubation such as during movement, patient transport, or extension of the head and neck.²⁰⁶ Proactive nursing care aimed at avoiding excessively long periods of intubation provides a protective benefit.³⁸⁸

Using Skilled Suctioning Techniques

Complications of tracheal suctioning during TLI include pain, coughing, hypoxemia, cardiac arrhythmias, bronchoconstriction, and mucosal damage. Consequently, it is important to suction the trachea carefully and never routinely.

Multiuse closed-system tracheal suction catheters save time, eliminate the need to disconnect the patient from the ventilator, and are preferable to open suctioning in patients with severe hypoxemia.³⁸⁹ There is, however, no consensus that they should be used to prevent VAP.³⁰⁴

Open suctioning of the trachea should be performed with clean or sterile gloves, a sterile, single-use catheter, and sterile fluid.³⁰⁴ Tracheal suctioning should ideally be performed by two attendants so as to avoid excessive movement of the ETT during suctioning and to maintain adequate oxygenation and ventilation. The patient should be adequately oxygenated and ventilated and suctioning intervals limited to a maximum of 10 to 15 seconds.

Avoiding Transmission of Mechanical Forces from the Endotracheal Tube Directly to the Patient

Elastic connectors and swivels between ventilator tubing and the ETT should be used. The weight of ventilator tubing should be supported by a folded towel on the patient's chest or by overhead supports. The ventilator tubing should not be tied to the bedrails. An ETT holder with integrated bite-block can help secure the position of the ETT and prevent the patient from biting and occluding the tube.

Minimizing Abrasion of the Laryngeal and Tracheal Mucosa

Any motion of the ETT during TLI applies shearing forces to the mucous membranes of the mouth, and more importantly, the glottis and trachea, promoting airway injury. Attempts at talking, swallowing, and moving by the patient should be discouraged. Coughing should be controlled and aggressive measures taken to manage agitation, seizures, and rigid posturing. The ventilator tubing, ETT, and patient should be thought of as one unit and moved together in order to minimize movement at the interface between the ETT and the patient's airway. Nasogastric tubes should be used only if necessary.

Miscellaneous Preventive Strategies

The ETT should be attached to a source of warm, fully humidified gas to promote clearing of airway secretions and to avoid desiccation of the tracheobronchial mucosa. If a heat and moisture exchanger is used, especially for prolonged periods of time, close observation of ETT patency and airway resistance is necessary because of the risk of tube occlusion. The patient should be reassured and given emotional support throughout the period of TLI. This may help avoid complications such as self-extubation and those caused by excessive motion and agitation. Because ventilated patients experience a high level of frustration and stress related to their difficulty in communication,³⁹⁰ care providers need to be especially attentive to patients' communication needs. Finally, the ETT should of course be removed as quickly as possible.

Modifying the Design of the Endotracheal Tube

Investigators have attempted to reduce the rate of complications of TLI by modifying the design of the ETT. Such innovations have led to the introduction into clinical practice of devices such as the foam-cuff ETT and the ETT with a suctioning lumen above the cuff. The foam-cuff tube described by Kamen and Wilkinson reduces lateral tracheal wall pressure to less than 15 mm Hg.³⁹¹ Experimental innovations have included shaping the ETT to match the anatomic contour of the airway,³⁹² using a balloon on the shaft of the ETT,^{126,261} covering the ETT shaft with foam,³¹⁶ lengthening the cuff to provide a larger contact area on the tracheal mucosa,³⁹³ using two cuffs,³⁹⁴ and adjusting the cuff design to prevent leakage of fluid into the lungs.³⁹⁵ Self-regulating devices to control cuff pressure have been designed,²⁵⁵ but they have not gained widespread acceptance.

Complications during and after Extubation

Respiratory care techniques to avoid complications of extubation are reviewed elsewhere.^{185,396} The patient should be adequately oxygenated and ventilated before extubation. The cuff should be fully deflated before the ETT is removed, and the pool of secretions above the cuff should be suctioned from the trachea immediately upon cuff deflation. The rate of failed extubations was reduced by a quality improvement program that addressed risk factors for failed extubation.³⁹⁷

Avoiding Stridor

Patients with a high risk of postextubation stridor (female gender, prolonged TLI, smaller leak volume on cuff-leak test) should be considered as candidates for prophylactic administration of corticosteroids to prevent or attenuate laryngeal edema. Older studies showed little or no benefit of single doses of corticosteroids to prevent laryngeal edema and postextubation stridor,^{398,399} but several recent reviews and meta-analyses indicate a clear benefit of multiple doses of corticosteroids administered 12 to 24 hours before extubation in reducing the occurrence of stridor and the need for reintubation.^{400–402}

Patients who have displayed one episode of postextubation stridor requiring reintubation represent a serious therapeutic challenge for subsequent attempts at extubation. In this situation, because no particular therapeutic regimen has been studied carefully in adults and proven to be of benefit, all measures are empiric. No matter what measures are employed, experience indicates the need for repeated intubation of such patients. In one study, such patients required a mean of 2.9 intubations.²²⁴ A trial of parenteral corticosteroids should be considered, but no controlled studies of this approach in adults have been reported. Nebulized epinephrine and helium–oxygen gas inhalation are also reasonable approaches. Weymuller et al¹⁰⁸ proposed a rigorous regimen in anticipation of a second episode of postextubation stridor, including general anesthesia in the operating room, followed by direct laryngoscopy. Deeb et al recommended immediate telarlaryngoscopy for all patients who appear to require tracheotomy because of failed extubation.¹⁴³

Important Unknowns

In spite of accumulated experience in airway management of critically ill patients spanning six decades, there is still much that is unknown about complications of TLI and their avoidance. Considering the major advances of the last few decades, it is reasonable to expect even more advances in the decades ahead. It is likely that some of the following questions will be addressed:

1. Are there better techniques to assure correct ETT placement into the trachea?
2. Can subsets of patients requiring TLI who would benefit from early tracheotomy be identified?
3. Is there an optimal size of ETT for individual patients based on gender and height?
4. Is there a better shape for the ETT to avoid posterior laryngeal injury?
5. Is abrasion of the mucosa as a result of motion between the patient and the ETT at the level of the posterior glottis an important risk factor for posterior glottis injury?
6. Would reducing ETT shaft stiffness decrease the risk of posterior glottis injury?

7. Is there a better cuff design to minimize tracheal surface injury?
8. What is the appropriate balance between cuff inflation pressure and airway peak inflation pressure, considering the competing needs of preventing tracheal mucosal ischemic injury and reducing leakage of secretions from above the cuff?
9. What is the optimal composition of the ETT shaft in relationship to its wall thickness, compliance, inner diameter, pressure-flow relationship, and biofilm accumulation?
10. What is the appropriate level of training and experience permissible to perform TLI?

The Future

Extensive experience with TLI over the last six decades has provided a huge database of ETT-related complications, mechanisms of airway injury, recommendations for management of the intubated patient, and strategies for prevention, early recognition, and treatment of the complications. Large prospective studies of complications of TLI provide special insight into the true frequency of complications and the related risk factors. Retrospective observations, individual case reports, and small case series, however, have also added to our growing knowledge about airway complications of TLI. The expanding literature has allowed us to record and classify seemingly countless examples of airway injury from TLI as well as tracheotomy. Since the development of high-volume, low-pressure ETT cuffs more than 40 years ago, however, major advances to prevent complications of TLI in critical care have not been forthcoming.

Opportunities to reduce the impact of complications of TLI include the following:

1. Performing large multicenter prospective studies of complications of TLI in critically ill adults and children to evaluate the optimal timing of tracheotomy in subsets of critically ill patients.
2. Promoting training and credentialing in advanced airway management, including management of the difficult airway, for personnel involved in critical care at all levels—paramedical personnel, emergency department staff, and critical care unit practitioners.
3. Supporting research in new design and composition of artificial airways, including research to make ETTs more compliant, especially at the level of the posterior larynx, to design new and safer cuffs, and to reduce biofilm on tube surfaces.
4. Performing further prospective studies of the use of alternatives to TLI, including the tracheoesophageal Combitube, the laryngeal-mask airway, and noninvasive positive airway pressure devices.
5. Developing clinical practice guidelines for managing TLI in critically ill patients.

Summary and Conclusions

Complications of TLI in critically ill adults are reviewed and classified according to the time period in which they occur: during ETT placement, while the ETT is in place, and during and after extubation. This chapter reviewed mechanisms of these adverse events and described steps to recognize, manage, and prevent complications of TLI. As the literature on complications of TLI expands, new insights to improve airway management of the critically ill patient continue to develop and evolve.

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